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Namespace MGeLabs.Utils.Attributes

Classes

[ReadOnlyFieldAttribute](#)

Attribute that marks a field as read-only in the Unity Inspector. Apply this attribute to a field to render it disabled (non-editable) in the editor.

Class ReadOnlyFieldAttribute

Namespace: MGeLabs.Utils.[Attributes](#)

Assembly: cs.temp.dll.dll

Attribute that marks a field as read-only in the Unity Inspector. Apply this attribute to a field to render it disabled (non-editable) in the editor.

```
public class ReadOnlyFieldAttribute : PropertyAttribute
```

Inheritance

object ← ReadOnlyFieldAttribute

Namespace MGeLabs.Utils.Data

Classes

[Batcher](#)

Provides utility methods for processing items in batches with support for callbacks and error handling.

[Crypter](#)

Provides methods for encrypting and decrypting strings using a simple XOR-based algorithm.

[ExternalAudioLoader](#)

A concrete implementation of [ExternalFileLoaderBase<T>](#) for loading audio files into objects.

[ExternalFileLoaderBase<T>](#)

Abstract base class for loading external files of type `T`. Can be extended to support different file types by implementing the abstract methods.

[ExternalTextureLoader](#)

A concrete implementation of [ExternalFileLoaderBase<T>](#) for loading texture files into objects.

[FadeTransitions](#)

Utility class for different transitions of various types.

[FileHandle](#)

Represents a file handle that provides methods for file operations such as reading, writing, and deleting files.

[FileHandleComparer](#)

Provides methods to compare two [FileHandle](#) objects for equality based on their full paths.

[FileManager](#)

Provides utility methods for managing files and directories within different storage locations.

[Formatters](#)

Provides utility methods for formatting values.

[JsonObject<T>](#)

Represents a generic JSON object that can be serialized and deserialized using Unity's `JsonUtility`. The derived class is required to have the `[Serializable]` attribute.

[LerpValue<T>](#)

A generic class for managing and interpolating a value towards a target value over time.

[ObservableField<T>](#)

Represents an observable field that notifies subscribers when its value changes.

[ObservableList<T>](#)

Represents a list that notifies subscribers when items are added or removed.

[WaitHelper](#)

Enums

[EStorageLocation](#)

Specifies the storage location for files within the application.

Class Batcher

Namespace: MGeLabs.Utils.[Data](#)

Assembly: cs.temp.dll.dll

Provides utility methods for processing items in batches with support for callbacks and error handling.

```
public static class Batcher
```

Inheritance

object ← Batcher

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ProcessInBatches<T>(IReadOnlyList<T>, Action<T>, int, float,
Action<T>, Action<T, bool>, Action<IReadOnlyList<T>>,
Action<IReadOnlyList<T>>, Action<IReadOnlyList<T>>,
Action<T, Exception>)

Executes an action on a list of items in batches, with optional callbacks for various stages of processing.

```
public static IEnumerable ProcessInBatches<T>(IReadOnlyList<T> itemsList, Action<T>  
actionOnItem, int itemsPerBatch = 5, float batchDelay = 0.1, Action<T> onItemProcessStart =  
null, Action<T, bool> onItemProcessFinished = null, Action<IReadOnlyList<T>> onBatchStart =  
null, Action<IReadOnlyList<T>> onBatchFinished = null, Action<IReadOnlyList<T>> onFinished =  
null, Action<T, Exception> onError = null)
```

Parameters

itemsList IReadOnlyList<T>

The list of items to process.

actionOnItem Action<T>

The action to execute on each item.

itemsPerBatch int

The number of items to process per batch. Defaults to 5.

batchDelay float

The delay (in seconds) between processing batches. Defaults to 0.1f.

onItemProcessStart Action<T>

Optional callback invoked before processing each item; receives the item as **T**.

onItemProcessFinished Action<T, bool>

Optional callback invoked after processing each item; receives the item as **T** and a **bool** indicating success.

onBatchStart Action<IReadOnlyList<T>>

Optional callback invoked before processing each batch; receives the batch as **IReadOnlyList<T>**.

onBatchFinished Action<IReadOnlyList<T>>

Optional callback invoked when a batch is completed; receives the batch as **IReadOnlyList<T>**.

onFinished Action<IReadOnlyList<T>>

Optional callback invoked when all items have been processed; receives the full list of processed items as **IReadOnlyList<T>**.

onError Action<T, Exception>

Optional callback invoked when an error occurs during item processing; receives the item that failed (**T**) and the thrown **Exception**.

Returns

IEnumerator

An enumerator that can be used to execute the batches over time.

Type Parameters

T

The type of items in the list.

Examples

```
ReadOnlyList<int> numbers = new ReadOnlyList<int>() { 1, 2, 3, 4, 5, 6, 7 };

StartCoroutine(Batcher.ProcessInBatches(
    numbers,
    n => Debug.Log($"Processing: {n}"),
    itemsPerBatch: 3,
    batchDelay: 0.5f,
    onBatchStart: batch => Debug.Log($"Batch starting: {string.Join(", ", batch)}"),
    onBatchFinished: batch => Debug.Log($"Batch finished: {string.Join(", ", batch)}"),
    onFinish: all => Debug.Log("All items processed.")
));
```

Exceptions

System.ArgumentNullException

Thrown if `actionOnItem` is null.

ProcessInBatches<T>(ReadOnlyList<T>, Func<T, IEnumerator>, int, float, Action<T>, Action<T, bool>, Action<ReadOnlyList<T>>, Action<ReadOnlyList<T>>, Action<ReadOnlyList<T>>)

Processes a list of items in batches, where each item is processed using a coroutine (IEnumerator).

```
public static IEnumerator ProcessInBatches<T>(ReadOnlyList<T> itemsList, Func<T,
IEnumerator> routineOnItem, int itemsPerBatch = 5, float batchDelay = 0.1, Action<T>
onItemProcessStart = null, Action<T, bool> onItemProcessFinished = null,
Action<ReadOnlyList<T>> onBatchStart = null, Action<ReadOnlyList<T>> onBatchFinished =
null, Action<ReadOnlyList<T>> onFinish = null)
```

Parameters

itemsList ReadOnlyList<T>

The list of items to process.

routineOnItem Func<T, IEnumerator>

The coroutine to execute for each item.

itemsPerBatch int

The number of items to process per batch. Defaults to 5.

batchDelay float

The delay (in seconds) between processing batches. Defaults to 0.1f.

onItemProcessStart Action<T>

Optional callback invoked before processing each item; receives the item as **T**.

onItemProcessFinished Action<T, bool>

Optional callback invoked after processing each item; receives the item as **T** and a **bool** indicating success.

onBatchStart Action<IReadOnlyList<T>>

Optional callback invoked before processing each batch; receives the batch as **IReadOnlyList<T>**.

onBatchFinished Action<IReadOnlyList<T>>

Optional callback invoked when a batch is completed; receives the batch as **IReadOnlyList<T>**.

onFinished Action<IReadOnlyList<T>>

Optional callback invoked when all items have been processed; receives the full list of processed items as **IReadOnlyList<T>**.

Returns

IEnumerator

An enumerator that can be used to execute the batches over time.

Type Parameters

T

The type of items in the list.

Examples

```
public class AudioLoaderExample : MonoBehaviour
{
    void Start()
    {
        List<int> numbers = new List<int> { 1, 2, 3, 4, 5, 6, 7 };
        StartCoroutine(Batcher.ProcessInBatches(
            numbers,
            number => ProcessNumberCoroutine(number),
            itemsPerBatch: 3,
            batchDelay: 0.5f,
            onBatchStart: batch => Debug.Log($"Batch starting: {string.Join("\", batch)})",
            onBatchFinished: batch => Debug.Log($"Batch finished: {string.Join("\", batch)})",
            onFinish: all => Debug.Log("All items processed")
        ));
    }

    private IEnumerator ProcessNumberCoroutine(int number)
    {
        Debug.Log($"Processing {number}");
        // Simulate async work
        yield return null;
    }
}
```

Exceptions

System.ArgumentNullException

Thrown if `routineOnItem` is null.

ProcessPreBatched<T>(IEnumerable<IEnumerable<T>>, Action<T>, float, Action<T>, Action<T, bool>, Action<IEnumerable<T>>, Action<IEnumerable<T>>, Action<IEnumerable<T>>, Action<T, Exception>)

Executes an action on pre-batched lists of items, with optional callbacks for various stages of processing.

```
public static IEnumerable ProcessPreBatched<T>(IReadOnlyList<IReadOnlyList<T>>  
preBatchedLists, Action<T> actionOnItem, float batchDelay = 0.1, Action<T>  
onItemProcessStart = null, Action<T, bool> onItemProcessFinished = null,  
Action<IReadOnlyList<T>> onBatchStart = null, Action<IReadOnlyList<T>> onBatchFinished =  
null, Action<IReadOnlyList<T>> onFinished = null, Action<T, Exception> onError = null)
```

Parameters

preBatchedLists IReadOnlyList<IReadOnlyList<T>>

The pre-batched lists of items to process.

actionOnItem Action<T>

The action to execute on each item.

batchDelay float

The delay (in seconds) between processing batches. Defaults to 0.1f.

onItemProcessStart Action<T>

Optional callback invoked before processing each item; receives the item as **T**.

onItemProcessFinished Action<T, bool>

Optional callback invoked after processing each item; receives the item as **T** and a **bool** indicating success.

onBatchStart Action<IReadOnlyList<T>>

Optional callback invoked before processing each batch; receives the batch as **IReadOnlyList<T>**.

onBatchFinished Action<IReadOnlyList<T>>

Optional callback invoked when a batch is completed; receives the batch as **IReadOnlyList<T>**.

onFinished Action<IReadOnlyList<T>>

Optional callback invoked when all items have been processed; receives the full list of processed items as **IReadOnlyList<T>**.

onError Action<T, Exception>

Optional callback invoked when an error occurs during item processing; receives the item that failed (T) and the thrown `Exception`.

Returns

IEnumerator

An enumerator that can be used to execute the batches over time.

Type Parameters

T

The type of items in the batches.

Examples

```
ReadOnlyList<ReadOnlyList<int>> batches = new ReadOnlyList<ReadOnlyList<int>>()
{
    new ReadOnlyList<int>() { 1, 2, 3 },
    new ReadOnlyList<int>() { 4, 5, 6 },
    new ReadOnlyList<int>() { 7 }
};

StartCoroutine(Batcher.ProcessPreBatched(
    batches,
    n => Debug.Log($"Processing: {n}"),
    batchDelay: 0.5f,
    onBatchStart: batch => Debug.Log($"Batch starting: {string.Join(", ", batch)}"),
    onBatchFinished: batch => Debug.Log($"Batch finished: {string.Join(", ", batch)}"),
    onFinish: all => Debug.Log("All items processed.")
));
```

Exceptions

System.ArgumentNullException

Thrown if `actionOnItem` is null.

`ProcessPreBatched<T>(ReadOnlyList<ReadOnlyList<T>>, Func<T, IEnumerator>, float, Action<T>, Action<T, bool>,`

Action<IReadOnlyList<T>>, Action<IReadOnlyList<T>>, Action<IReadOnlyList<T>>)

Processes pre-batched lists of items, where each item is processed using a coroutine (IEnumerator).

```
public static IEnumerator ProcessPreBatched<T>(IReadOnlyList<IReadOnlyList<T>>  
preBatchedLists, Func<T, IEnumerator> routineOnItem, float batchDelay = 0.1, Action<T>  
onItemProcessStart = null, Action<T, bool> onItemProcessFinished = null,  
Action<IReadOnlyList<T>> onBatchStart = null, Action<IReadOnlyList<T>> onBatchFinished =  
null, Action<IReadOnlyList<T>> onFinished = null)
```

Parameters

preBatchedLists IReadOnlyList<IReadOnlyList<T>>

The pre-batched lists of items to process.

routineOnItem Func<T, IEnumerator>

The coroutine to execute for each item.

batchDelay float

The delay (in seconds) between processing batches. Defaults to 0.1f.

onItemProcessStart Action<T>

Optional callback invoked before processing each item; receives the item as **T**.

onItemProcessFinished Action<T, bool>

Optional callback invoked after processing each item; receives the item as **T** and a **bool** indicating success.

onBatchStart Action<IReadOnlyList<T>>

Optional callback invoked before processing each batch; receives the batch as **IReadOnlyList<T>**.

onBatchFinished Action<IReadOnlyList<T>>

Optional callback invoked when a batch is completed; receives the batch as **IReadOnlyList<T>**.

onFinished Action<IReadOnlyList<T>>

Optional callback invoked when all items have been processed; receives the full list of processed items as `IReadOnlyList<T>`.

Returns

IEnumerator

An enumerator that can be used to execute the batches over time.

Type Parameters

T

The type of items in the batches.

Exceptions

System.ArgumentNullException

Thrown if `routineOnItem` is null.

Class Crypter

Namespace: MGeLabs.Utils.[Data](#)

Assembly: cs.temp.dll.dll

Provides methods for encrypting and decrypting strings using a simple XOR-based algorithm.

```
public class Crypter
```

Inheritance

object ← Crypter

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Decrypt(byte[], string)

Decrypts a byte array using a XOR-based algorithm and returns the decrypted string.

```
public static string Decrypt(byte[] encryptedBytes, string encryptionKey)
```

Parameters

encryptedBytes byte[]

The byte array containing the encrypted data.

encryptionKey string

The encryption key used for the XOR operation.

Returns

string

The decrypted string.

Examples

```
string text = "HelloWorld";  
string key = "k";  
byte[] encrypted = Crypter.Encrypt(text, key);  
string decrypted = Crypter.Decrypt(encrypted, key);  
Debug.Log(decrypted); // "HelloWorld"
```

DecryptFromBase64(string, string)

Decrypts a Base64-encoded string using a XOR-based algorithm and returns the decrypted string.

```
public static string DecryptFromBase64(string encryptedBase64, string encryptionKey)
```

Parameters

encryptedBase64 string

The Base64-encoded string containing the encrypted data.

encryptionKey string

The encryption key used for the XOR operation.

Returns

string

The decrypted string.

Examples

```
string key = "secret";  
string encrypted = Crypter.EncryptToBase64("HiddenMessage", key);  
string decrypted = Crypter.DecryptFromBase64(encrypted, key);  
Debug.Log(decrypted); // Outputs: "HiddenMessage"
```

Encrypt(string, string)

Encrypts a string using a XOR-based algorithm and returns the encrypted data as a byte array.

```
public static byte[] Encrypt(string input, string encryptionKey)
```

Parameters

input string

The input string to encrypt.

encryptionKey string

The encryption key used for the XOR operation.

Returns

byte[]

A byte array containing the encrypted data.

Examples

```
string original = "Secret123";  
string key = "key!";  
byte[] encrypted = Crypter.Encrypt(original, key);  
// encrypted contains binary data
```

EncryptToBase64(string, string)

Encrypts a string using a XOR-based algorithm and returns the encrypted data as a Base64-encoded string.

```
public static string EncryptToBase64(string input, string encryptionKey)
```

Parameters

input string

The input string to encrypt.

encryptionKey string

The encryption key used for the XOR operation.

Returns

string

A Base64-encoded string containing the encrypted data.

Examples

```
string secret = "MyPassword";  
string key = "abc123";  
string encryptedBase64 = Crypter.EncryptToBase64(secret, key);  
Debug.Log(encryptedBase64); // Outputs something like: "GhoaFh1R..."
```

Enum EStorageLocation

Namespace: MGeLabs.Utils.[Data](#)

Assembly: cs.temp.dll.dll

Specifies the storage location for files within the application.

```
public enum EStorageLocation
```

Fields

DataPath = 1

(Application.dataPath) Represents the application's data path, often used for accessing files bundled with the application.

Persistent = 0

(Application.persistentDataPath) Represents the persistent storage location, typically used for saving data that should remain across application sessions.

StreamingAssets = 3

(Application.streamingAssetsPath) Represents the streaming assets folder, often used for accessing read-only data bundled with the application.

Temporary = 2

(Application.temporaryCachePath) Represents a temporary storage location, typically used for storing data that does not need to persist.

Class ExternalAudioLoader

Namespace: MGeLabs.Utils.[Data](#)

Assembly: cs.temp.dll.dll

A concrete implementation of [ExternalFileLoaderBase<T>](#) for loading audio files into objects.

```
public class ExternalAudioLoader : ExternalFileLoaderBase<AudioClip>
```

Inheritance

object ← [ExternalFileLoaderBase](#)<AudioClip> ← ExternalAudioLoader

Inherited Members

[ExternalFileLoaderBase<AudioClip>.LoadFromDirectory\(string, EStorageLocation, Action<IReadOnlyList<AudioClip>>, Action<IReadOnlyList<AudioClip>>, Action<FileHandle, int>, int\)](#) ,
[ExternalFileLoaderBase<AudioClip>.FindFilesInDirectory\(string, EStorageLocation\)](#) ,
object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

SupportedExtensions

Gets the list of supported audio file extensions.

```
protected override List<string> SupportedExtensions { get; }
```

Property Value

List<string>

Methods

GetAudioType(FileHandle)

Determines the audio type based on the file extension.

```
protected virtual AudioType GetAudioType(FileHandle fileHandle)
```

Parameters

fileHandle [FileHandle](#)

The file handle representing the audio file.

Returns

AudioType

The corresponding .

LoadSingleFileRoutine(FileHandle, Action<AudioClip>, Action<FileHandle, int>)

Loads a single audio file asynchronously.

```
protected override IEnumerator LoadSingleFileRoutine(FileHandle handle, Action<AudioClip>  
onLoaded, Action<FileHandle, int> onError = null)
```

Parameters

handle [FileHandle](#)

The file handle representing the audio file to load.

onLoaded Action<AudioClip>

Callback invoked with the loaded audio clip upon success.

onError Action<[FileHandle](#), int>

Callback invoked if the audio file fails to load.

Returns

IEnumerator

An enumerator for the coroutine.

Class ExternalFileLoaderBase<T>

Namespace: MGeLabs.Utils.[Data](#)

Assembly: cs.temp.dll.dll

Abstract base class for loading external files of type **T**. Can be extended to support different file types by implementing the abstract methods.

```
public abstract class ExternalFileLoaderBase<T>
```

Type Parameters

T

The type of data to be returned (AudioClip, VideoClip, ...).

Inheritance

object $\leftarrow$ ExternalFileLoaderBase<T>

Derived

[ExternalAudioLoader](#), [ExternalTextureLoader](#)

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

SupportedExtensions

Gets the list of supported file extensions for this loader.

```
protected abstract List<string> SupportedExtensions { get; }
```

Property Value

List<string>

Methods

FindFilesInDirectory(string, EStorageLocation)

Finds files in the specified directory that match the supported extensions.

```
protected virtual List<FileHandle> FindFilesInDirectory(string relativePath,  
EStorageLocation location = EStorageLocation.DataPath)
```

Parameters

relativePath string

The relative path to the directory.

location [EStorageLocation](#)

The storage location of the directory (default is DataPath).

Returns

List<[FileHandle](#)>

A list of file handles for the matching files.

LoadFromDirectory(string, EStorageLocation, Action<IReadOnlyList<T>>, Action<IReadOnlyList<T>>, Action<FileHandle, int>, int)

Loads files from a specified directory asynchronously in batches.

```
public virtual IEnumerator LoadFromDirectory(string relativePath, EStorageLocation location  
= EStorageLocation.DataPath, Action<IReadOnlyList<T>> onLoadingFinished = null,  
Action<IReadOnlyList<T>> onBatchFinished = null, Action<FileHandle, int> onError = null, int  
loadBatchSize = 10)
```

Parameters

relativePath string

The relative path to the directory containing the files.

location [EStorageLocation](#)

The storage location of the directory (default is DataPath).

onLoadingFinished Action<IReadOnlyList<T>>

Callback invoked when all files have been loaded successfully; receives the list of loaded items.

onBatchFinished Action<IReadOnlyList<T>>

Callback invoked after each batch of files is loaded; receives the full list of currently loaded items.

onError Action<[FileHandle](#), int>

Callback invoked for each file that fails to load; receives the file handle and an integer error code.

loadBatchSize int

The number of files to load per batch (default is 10).

Returns

IEnumerator

An enumerator for the coroutine.

LoadSingleFileRoutine(FileHandle, Action<T>, Action<FileHandle, int>)

Loads a single file asynchronously. Must be implemented by derived classes to handle the loading/parsing of the specific file type.

```
protected abstract IEnumerator LoadSingleFileRoutine(FileHandle handle, Action<T>  
onLoadingFinished, Action<FileHandle, int> onError = null)
```

Parameters

handle [FileHandle](#)

The file handle representing the file to load.

onLoadingFinished Action<T>

Callback invoked with the loaded item upon success.

onError Action<[FileHandle](#), int>

Callback invoked if the file fails to load.

Returns

IEnumerator

An enumerator for the coroutine.

Class ExternalTextureLoader

Namespace: MGeLabs.Utils.[Data](#)

Assembly: cs.temp.dll.dll

A concrete implementation of [ExternalFileLoaderBase<T>](#) for loading texture files into objects.

```
public class ExternalTextureLoader : ExternalFileLoaderBase<Texture2D>
```

Inheritance

object ← [ExternalFileLoaderBase](#)<Texture2D> ← ExternalTextureLoader

Inherited Members

[ExternalFileLoaderBase<Texture2D>.LoadFromDirectory\(string, EStorageLocation, Action<IReadOnlyList<Texture2D>>, Action<IReadOnlyList<Texture2D>>, Action<FileHandle, int>, int\)](#), [ExternalFileLoaderBase<Texture2D>.FindFilesInDirectory\(string, EStorageLocation\)](#), object.Equals(object), object.Equals(object, object), object.GetHashCode(), object.GetType(), object.MemberwiseClone(), object.ReferenceEquals(object, object), object.ToString()

Properties

SupportedExtensions

Supported texture file extensions.

```
protected override List<string> SupportedExtensions { get; }
```

Property Value

List<string>

Methods

LoadSingleFileRoutine(FileHandle, Action<Texture2D>, Action<FileHandle, int>)

Loads a single texture file asynchronously.

```
protected override IEnumerator LoadSingleFileRoutine(FileHandle handle, Action<Texture2D>  
onLoadingFinished, Action<FileHandle, int> onError = null)
```

Parameters

handle [FileHandle](#)

The file handle representing the texture file.

onLoadingFinished Action<Texture2D>

Callback invoked with the loaded Texture2D upon success.

onError Action<[FileHandle](#), int>

Callback invoked if the texture fails to load.

Returns

IEnumerator

An enumerator for the coroutine.

Class FadeTransitions

Namespace: MGeLabs.Utils.[Data](#)

Assembly: cs.temp.dll.dll

Utility class for different transitions of various types.

```
public class FadeTransitions : MonoBehaviour
```

Inheritance

object ← FadeTransitions

Methods

FadeLightColor(Light, Color, float)

Coroutine to fade a light's color over time.

```
public static IEnumerator FadeLightColor(Light targetLight, Color targetColor, float  
duration = 1)
```

Parameters

targetLight Light

The light to modify.

targetColor Color

The target color to fade to.

duration float

The duration of the fade in seconds.

Returns

IEnumerator

An IEnumerator for use in a coroutine.

Examples

```
// Example: Fade a light's color to blue over 2 seconds.  
StartCoroutine(FadeTransitions.FadeLightColor(  
    targetLight: myLight,  
    targetColor: Color.blue,  
    duration: 2f  
));
```

FadeLightIntensity(Light, float, float)

Coroutine to fade a light's intensity over time.

```
public static IEnumerator FadeLightIntensity(Light targetLight, float targetIntensity, float  
duration = 1)
```

Parameters

targetLight Light

The light to modify.

targetIntensity float

The target intensity to fade to.

duration float

The duration of the fade in seconds.

Returns

IEnumerator

An IEnumerator for use in a coroutine.

Examples

```
// Example: Fade a light's intensity to 0 over 1.5 seconds.  
StartCoroutine(FadeTransitions.FadeLightIntensity(  
    targetLight: myLight,  
    targetIntensity: 0f,
```

```
        duration: 1.5f  
    ));
```

FadeMaterialColor(Material, Color, string, int, float)

Coroutine to fade a material's color property over time.

```
public static IEnumerator FadeMaterialColor(Material targetMaterial, Color targetColor,  
string fieldName = "_EmissionColor", int materialPropertyId = -1, float duration = 1)
```

Parameters

targetMaterial Material

The material to modify.

targetColor Color

The target color to fade to.

fieldName string

The name of the color property to modify (default is "_EmissionColor").

materialPropertyId int

Optional property ID for the color property. If provided, it overrides fieldName.

duration float

The duration of the fade in seconds.

Returns

IEnumerator

An IEnumerator for use in a coroutine.

Examples

```
// Example: Fade a material's emission color to red over 3 seconds.  
StartCoroutine(FadeTransitions.FadeMaterialColor(  
    targetMaterial, Color.red, 3f));
```

```
targetMaterial: myRenderer.material,  
targetColor: Color.red,  
fieldName: "_EmissionColor",  
duration: 3f  
));
```

FadeValue<T>(Func<T>, Action<T>, T, Func<T, T, float, T>, float)

Generic coroutine to fade a value over time.

```
public static IEnumerator FadeValue<T>(Func<T> getter, Action<T> setter, T targetValue,  
Func<T, T, float, T> lerpFunc, float duration = 1)
```

Parameters

getter Func<T>

Function to get the current value.

setter Action<T>

Action to set the new value.

targetValue T

The target value to fade to.

lerpFunc Func<T, T, float, T>

Function to interpolate between values.

duration float

The duration of the fade in seconds.

Returns

IEnumerator

An IEnumerator for use in a coroutine.

Type Parameters

T

The type of the value to fade.

Examples

```
// Example: Fade a float value from 0 to 1 over 2 seconds.  
StartCoroutine(FadeTransitions.FadeValue(  
    getter: () => someValue,  
    setter: val => someValue = val,  
    targetValue: 1f,  
    lerpFunc: Mathf.Lerp,  
    duration: 2f  
));
```

Class FileHandle

Namespace: MGeLabs.Utils.[Data](#)

Assembly: cs.temp.dll.dll

Represents a file handle that provides methods for file operations such as reading, writing, and deleting files.

```
public class FileHandle
```

Inheritance

object ← FileHandle

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

FileHandle(string, EStorageLocation)

Initializes a new instance of the [FileHandle](#) class with the specified relative path and storage location. Ensures the directory for the file exists.

```
public FileHandle(string relativePath, EStorageLocation location  
    = EStorageLocation.Persistent)
```

Parameters

relativePath string

The relative path of the file.

location [EStorageLocation](#)

The storage location. Defaults to [Persistent](#).

Properties

Exists

```
public virtual bool Exists { get; }
```

Property Value

bool

Returns **true** if the file exists; otherwise, **false**.

Extension

```
public string Extension { get; }
```

Property Value

string

Returns the file extension.

FileName

```
public virtual string FileName { get; }
```

Property Value

string

The name of the file without the directory path.

FileNameWithoutExtension

```
public virtual string FileNameWithoutExtension { get; }
```

Property Value

string

The name of the file without the directory path and file extension.

FullPath

```
public virtual string FullPath { get; }
```

Property Value

string

The full path to the file, including the directory and file name.

IsReadOnly

```
public virtual bool IsReadOnly { get; }
```

Property Value

bool

Returns **true** if the file is read-only; otherwise, **false**.

Location

```
public virtual EStorageLocation Location { get; }
```

Property Value

[EStorageLocation](#)

The storage location of the file, such as [Persistent](#).

RelativePath

```
public virtual string RelativePath { get; }
```

Property Value

string

The relative path of the file from the root of the specified storage location.

Methods

AppendText(string)

Appends the specified text content to the file.

```
public virtual bool AppendText(string content)
```

Parameters

content string

The text content to append to the file.

Returns

bool

true if the operation was successful; otherwise, **false**.

Examples

```
FileHandle handle = new FileHandle("log.txt");  
handle.AppendText("Log entry\n");
```

Exceptions

System.IO.IOException

Thrown if appending to the file fails.

Copy(string, EStorageLocation?, bool)

Copies the current file to a new location and returns a new [FileHandle](#) instance for the copied file.

```
public virtual FileHandle Copy(string newRelativePath, EStorageLocation? location = null,
bool canOverwrite = false)
```

Parameters

newRelativePath string

The relative path for the new file location.

location [EStorageLocation?](#)

The storage location for the new file. If **null**, the current file's location is used.

canOverwrite bool

If **true**, allows overwriting an existing file at the new location; otherwise, throws an error if the target file already exists.

Returns

[FileHandle](#)

A new [FileHandle](#) instance representing the copied file, or **null** if the operation fails.

Examples

```
FileHandle file = new FileHandle("data/source.txt");
FileHandle copiedFile = file.Copy("data/destination.txt", canOverwrite: true);
```

Exceptions

System.ArgumentException

Thrown if the new relative path is invalid.

System.IO.FileNotFoundException

Thrown if the file does not exist.

System.IO.IOException

Thrown if writing to the file fails.

Delete()

Deletes the current file.

```
public virtual bool Delete()
```

Returns

bool

`true` if the file was successfully deleted; otherwise, `false`.

Examples

```
FileHandle handle = new FileHandle("temp/data.txt");  
handle.Delete();
```

Exceptions

System.IO.IOException

Thrown if the delete operation fails.

Dump()

Logs the contents of the file to the Unity console.

```
public void Dump()
```

Examples

```
FileHandle handle = new FileHandle("logs/output.txt");  
handle.Dump(); // Logs file contents to Unity console.
```

FileSizeBytes()

```
public virtual long FileSizeBytes()
```

Returns

long

Returns the size of the file in bytes, or 0 if the file does not exist.

LastModified()

```
public virtual DateTime? LastModified()
```

Returns

DateTime?

Returns the last modified date and time of the file, or `null` if the file does not exist.

Move(string, EStorageLocation?, bool)

Moves the current file to a new location and returns a new [FileHandle](#) instance for the moved file.

```
public virtual FileHandle Move(string newRelativePath, EStorageLocation? location = null,  
bool canOverwrite = false)
```

Parameters

`newRelativePath` string

The relative path for the new file location.

location [EStorageLocation?](#)

canOverwrite bool

If **true**, allows overwriting an existing file at the new location; otherwise, throws an error if the target file already exists.

Returns

[FileHandle](#)

A new [FileHandle](#) instance representing the moved file, or **null** if the operation fails.

Examples

```
FileHandle file = new FileHandle("data/oldname.txt");  
FileHandle movedFile = file.Move("data/newname.txt", canOverwrite: true);
```

Exceptions

System.IO.FileNotFoundException

Thrown if the file does not exist.

System.ArgumentException

Thrown if the new relative path is invalid.

System.IO.IOException

Thrown if the target file already exists.

System.IO.IOException

Thrown if writing to the file fails.

OpenReadStream()

Opens and returns a read-only FileStream for this file.

```
public virtual FileStream OpenReadStream()
```

Returns

FileStream

A System.IO.FileStream opened for read.

Examples

```
FileHandle handle = new FileHandle("data.txt");
using (FileStream stream = handle.OpenReadStream())
{
    // Read from stream
}
```

Exceptions

System.IO.FileNotFoundException

Thrown if the file does not exist.

OpenWriteStream(bool)

Opens and returns a write-only FileStream for this file.

```
public virtual FileStream OpenWriteStream(bool overwrite = true)
```

Parameters

overwrite bool

If true, the file will be recreated; if false, data will be appended.

Returns

FileStream

A System.IO.FileStream opened for write.

Examples

```
FileHandle handle = new FileHandle("log.txt");
using (FileStream stream = handle.OpenWriteStream(overwrite: true))
{
    byte[] bytes = Encoding.UTF8.GetBytes("Hello");
    stream.Write(bytes, 0, bytes.Length);
}
```

ReadBytes()

Reads the content of the file as a byte array.

```
public virtual byte[] ReadBytes()
```

Returns

byte[]

The content of the file as a byte array, or `null` if the file does not exist.

Examples

```
FileHandle handle = new FileHandle("data.bin");
byte[] data = handle.ReadBytes();
```

Exceptions

System.IO.IOException

Thrown if the file cannot be read.

ReadText()

Reads the content of the file as a string.

```
public virtual string ReadText()
```

Returns

string

The content of the file as a string, or `null` if the file does not exist.

Examples

```
FileHandle handle = new FileHandle("log.txt");  
string content = handle.ReadText();  
Debug.Log(content);
```

Exceptions

System.IO.IOException

Thrown if the file cannot be read.

Rename(string, bool)

Renames the current file by moving it to a new name within the same directory.

```
public virtual FileHandle Rename(string newName, bool canOverwrite = false)
```

Parameters

`newName` string

The new name for the file.

`canOverwrite` bool

If `true`, allows overwriting an existing file with the new name; otherwise, throws an error if the target file already exists.

Returns

[FileHandle](#)

A new [FileHandle](#) instance representing the renamed file, or `null` if the operation fails.

ToString()

Returns the full path of the file as a string.

```
public override string ToString()
```

Returns

string

The full path of the file.

WriteBytes(byte[])

Writes the specified byte array content to the file, overwriting any existing content.

```
public virtual bool WriteBytes(byte[] content)
```

Parameters

content byte[]

The byte array content to write to the file.

Returns

bool

true if the operation was successful; otherwise, **false**.

Examples

```
FileHandle handle = new FileHandle("binary.bin");  
handle.WriteBytes(new byte[] { 1, 2, 3 });
```

Exceptions

System.IO.IOException

Thrown if writing to the file fails.

WriteText(string, Encoding)

Writes the specified text content to the file using the provided encoding, overwriting any existing content.

```
public virtual bool WriteText(string content, Encoding encoding)
```

Parameters

content string

The text content to write to the file.

encoding Encoding

The encoding to use when writing the text content.

Returns

bool

true if the operation was successful; otherwise, **false**.

Examples

```
FileHandle handle = new FileHandle("utf8.txt");  
handle.WriteText("Hello", Encoding.UTF8);
```

Exceptions

System.IO.IOException

Thrown if writing to the file fails.

WriteText(string)

Writes the specified text content to the file, overwriting any existing content.

```
public virtual bool WriteText(string content)
```

Parameters

content string

The text content to write to the file.

Returns

bool

true if the operation was successful; otherwise, **false**.

Examples

```
FileHandle handle = new FileHandle("output.txt");  
handle.WriteText("Hello, world!");
```

Exceptions

System.IO.IOException

Thrown if writing to the file fails.

Class FileHandleComparer

Namespace: MGeLabs.Utils.[Data](#)

Assembly: cs.temp.dll.dll

Provides methods to compare two [FileHandle](#) objects for equality based on their full paths.

```
public class FileHandleComparer : IEqualityComparer<FileHandle>
```

Inheritance

object ← FileHandleComparer

Methods

Equals(FileHandle, FileHandle)

Determines whether the specified [FileHandle](#) objects are equal by comparing their full paths.

```
public bool Equals(FileHandle x, FileHandle y)
```

Parameters

x [FileHandle](#)

The first [FileHandle](#) to compare.

y [FileHandle](#)

The second [FileHandle](#) to compare.

Returns

bool

true if the full paths of both [FileHandle](#) objects are equal; otherwise, **false**.

GetHashCode(FileHandle)

```
public int GetHashCode(FileHandle obj)
```

Parameters

obj [FileHandle](#)

Returns

int

Class FileManager

Namespace: MGeLabs.Utils.[Data](#)

Assembly: cs.temp.dll.dll

Provides utility methods for managing files and directories within different storage locations.

```
public static class FileManager
```

Inheritance

object ← FileManager

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ClearDirectory(string, EStorageLocation)

Deletes all contents of a directory without deleting the directory itself.

```
public static void ClearDirectory(string relativePath, EStorageLocation location  
= EStorageLocation.Persistent)
```

Parameters

relativePath string

The relative path of the directory to clear.

location [EStorageLocation](#)

The storage location. Defaults to [Persistent](#).

Examples

```
FileManager.ClearDirectory("Cache");
```

Exceptions

System.IO.IOException

Thrown if file or directory deletion fails.

System.UnauthorizedAccessException

Thrown if the operation lacks necessary permissions.

CopyDirectory(string, string, EStorageLocation)

Copies the contents of a source directory to a target directory within a specified storage location.

```
public static bool CopyDirectory(string sourceRelativePath, string targetRelativePath,
    EStorageLocation location = EStorageLocation.Persistent)
```

Parameters

sourceRelativePath string

The relative path of the source directory.

targetRelativePath string

The relative path of the target directory.

location [EStorageLocation](#)

The storage location to use. Defaults to [Persistent](#).

Returns

bool

true if the copy was successful; otherwise throws an exception.

Examples

```
FileManager.CopyDirectory("Configs", "Backup/Configs");
```

Exceptions

System.IO.DirectoryNotFoundException

Thrown if the source directory does not exist.

System.IO.IOException

Thrown if copying fails due to IO errors.

System.UnauthorizedAccessException

Thrown if the operation lacks necessary permissions.

DeleteDirectory(string, EStorageLocation)

Deletes a specified directory within a given storage location.

```
public static bool DeleteDirectory(string relativePath, EStorageLocation location  
= EStorageLocation.Persistent)
```

Parameters

relativePath string

The relative path of the directory to delete.

location [EStorageLocation](#)

The storage location to use for resolving the directory path. Defaults to [Persistent](#).

Returns

bool

true if the directory is deleted successfully; **false** if an error occurs.

Exceptions

System.IO.IOException

Thrown if file or directory deletion fails.

System.UnauthorizedAccessException

Thrown if the operation lacks necessary permissions.

EnsureDirectoryExists(string, EStorageLocation)

Ensures that a directory exists at the specified relative path within a given storage location. If the directory does not exist, it is created.

```
public static void EnsureDirectoryExists(string relativePath, EStorageLocation location  
= EStorageLocation.Persistent)
```

Parameters

relativePath string

The relative path of the directory to check or create.

location [EStorageLocation](#)

The storage location to use for resolving the directory path. Defaults to [Persistent](#).

GetFilesInDirectory(string, EStorageLocation, string)

Retrieves a list of files from a specified directory within a given storage location.

```
public static List<FileHandle> GetFilesInDirectory(string relativePath, EStorageLocation  
location = EStorageLocation.Persistent, string searchPattern = "*.*")
```

Parameters

relativePath string

The relative path to the directory from which to retrieve files.

location [EStorageLocation](#)

The storage location to use for resolving the directory path. Defaults to [Persistent](#).

searchPattern string

The search pattern to filter files. Defaults to "." (all files).

Returns

List<[FileHandle](#)>

A list of [FileHandle](#) objects representing the files in the specified directory. If the directory does not exist, an empty list is returned.

Examples

```
List<FileHandle> files = FileManager.GetFilesInDirectory("Logs", searchPattern: "*.log");
foreach (FileHandle file in files)
    file.Delete();
```

GetRootPath(EStorageLocation)

Retrieves the root path for the specified storage location.

```
public static string GetRootPath(EStorageLocation location)
```

Parameters

location [EStorageLocation](#)

The storage location for which to retrieve the root path.

Returns

string

The root path of the specified storage location.

GetSubdirectories(string, EStorageLocation)

Retrieves a list of subdirectories from a specified directory within a given storage location.

```
public static List<string> GetSubdirectories(string relativePath, EStorageLocation location
= EStorageLocation.Persistent)
```

Parameters

relativePath string

The relative path to the directory from which to retrieve subdirectories.

location [EStorageLocation](#)

The storage location to use for resolving the directory path. Defaults to [Persistent](#).

Returns

List<string>

A list of relative paths representing the subdirectories in the specified directory. If the directory does not exist, an empty list is returned.

Examples

```
List<string> subdirs = FileManager.GetSubdirectories("Projects");
foreach (string dir in subdirs)
    Debug.Log(dir);
```

MoveDirectory(string, string, EStorageLocation)

Moves a directory from one path to another within a given storage location.

```
public static bool MoveDirectory(string fromRelativePath, string toRelativePath,
    EStorageLocation location = EStorageLocation.Persistent)
```

Parameters

fromRelativePath string

The relative path of the source directory.

toRelativePath string

The relative path of the target directory.

location [EStorageLocation](#)

The storage location. Defaults to [Persistent](#).

Returns

bool

true if the move succeeds; otherwise throws an exception.

Examples

```
FileManager.MoveDirectory("TempData", "Archived/TempData");
```

Exceptions

System.IO.IOException

Thrown if the target directory already exists or if the move fails.

TryIOAction(Action, string, Action<Exception, string>)

Executes an I/O action and handles any exceptions that occur during its execution.

```
public static bool TryIOAction(Action action, string errorContext, Action<Exception, string>  
onError = null)
```

Parameters

action Action

The I/O action to execute.

errorContext string

A descriptive context for the error, used to provide additional information in the error message.

onError Action<Exception, string>

An optional callback to handle exceptions. The callback receives the exception and the formatted error message.

Returns

bool

`true` if the action executes successfully; `false` if an exception occurs.

Class Formatters

Namespace: MGeLabs.Utils.[Data](#)

Assembly: cs.temp.dll.dll

Provides utility methods for formatting values.

```
public static class Formatters
```

Inheritance

object ← Formatters

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

FormatDoubleToUsdString(double)

Formats a double value to a USD currency string.

```
public static string FormatDoubleToUsdString(double value)
```

Parameters

value double

The double value to format.

Returns

string

A formatted USD currency string.

Examples

```
double price = 49.99;  
string formatted = Formatters.FormatDoubleToUsdString(price);  
Debug.Log(formatted); // Output: "$49.99"
```

FormatFloatToTimeString(float)

Formats a float value representing time in seconds to a string in the format "MM:SS:FF".

```
public static string FormatFloatToTimeString(float time)
```

Parameters

time float

The time value in seconds.

Returns

string

A formatted time string in the format "MM:SS:FF".

Examples

```
float playTime = 125.37f; // 2 minutes, 5 seconds, 370 milliseconds  
string formatted = Formatters.FormatFloatToTimeString(playTime);  
Debug.Log(formatted); // Output: "02:05:37"
```

Class JsonObject<T>

Namespace: MGeLabs.Utils.[Data](#)

Assembly: cs.temp.dll.dll

Represents a generic JSON object that can be serialized and deserialized using Unity's JsonUtility. The derived class is required to have the [Serializable] attribute.

```
public abstract class JsonObject<T>
```

Type Parameters

T

The type of the object to be deserialized.

Inheritance

object ← JsonObject<T>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Methods

FromJson(string)

```
public static T FromJson(string json)
```

Parameters

json string

Returns

T

Examples


```
// Define a serializable data model
[Serializable]
public class PlayerData : JsonObject<PlayerData>
{
    public string playerName;
    public int score;
}

// Create an instance and serialize it
PlayerData data = new PlayerData
{
    playerName = "Alex",
    score = 150
};

string json = data.ToJson();
Debug.Log(json);
// Output: {"playerName":"Alex","score":150}

// You can also deserialize it back:
PlayerData deserialized = PlayerData.FromJson(json);
Debug.Log(deserialized.playerName);
// Output: Alex
```

ToJson(bool)

Serializes the current object to a JSON string.

```
public string ToJson(bool prettyPrint = false)
```

Parameters

prettyPrint bool

If true, formats the JSON string with better readability.

Returns

string

A JSON string representation of the current object.

Examples

```
[Serializable]
public class PlayerData : JsonObject<PlayerData>
{
    public string playerName;
    public int score;
}

PlayerData data = new PlayerData { playerName = "Alex", score = 150 };
string json = data.ToJson();
Debug.Log(json); // Output: {"playerName":"Alex","score":150}
```

ToString()

Overrides the ToString method to return the JSON representation of the object.

```
public override string ToString()
```

Returns

string

A JSON string representation of the current object.

Class LerpValue<T>

Namespace: MGeLabs.Utils.[Data](#)

Assembly: cs.temp.dll.dll

A generic class for managing and interpolating a value towards a target value over time.

```
public class LerpValue<T>
```

Type Parameters

T

The type of the value to interpolate.

Inheritance

object $\leftarrow$ LerpValue<T>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Examples

```
using UnityEngine;

public class ExampleUsage : MonoBehaviour
{
    private LerpValue<float> lerpAlpha;

    void Start()
    {
        // Start with 0, use Mathf.Lerp, and set speed to 2
        lerpAlpha = new LerpValue<float>(0f, Mathf.Lerp, 2f);
        lerpAlpha.SetTarget(1f); // Targeting full alpha
    }

    void Update()
    {
        lerpAlpha.Update(Time.deltaTime);
        float currentAlpha = lerpAlpha.Value;
        // Use currentAlpha for fading UI, etc.
    }
}
```

```
}  
}
```

Constructors

LerpValue(T, Func<T, T, float, T>, float)

Initializes a new instance of the [LerpValue<T>](#) class.

```
public LerpValue(T initialValue, Func<T, T, float, T> lerpFunction, float speed = 5)
```

Parameters

initialValue T

The initial value of the interpolation.

lerpFunction Func<T, T, float, T>

The interpolation function to use.

speed float

The speed of interpolation (default is 5f).

Exceptions

System.ArgumentNullException

Thrown if **lerpFunction** is null.

Fields

currentValue

The current interpolated value.

```
protected T currentValue
```

Field Value

T

lerpFunc

The interpolation function used to calculate the interpolated value.

```
protected readonly Func<T, T, float, T> lerpFunc
```

Field Value

Func<T, T, float, T>

lerpSpeed

The speed at which the value interpolates towards the target.

```
protected float lerpSpeed
```

Field Value

float

targetValue

The target value to interpolate towards.

```
protected T targetValue
```

Field Value

T

Properties

LerpSpeed

Gets the speed at which the value interpolates towards the target.

```
public virtual float LerpSpeed { get; }
```

Property Value

float

The speed of interpolation.

Target

Gets the target value.

```
public virtual T Target { get; }
```

Property Value

T

The target value.

Value

Gets the current interpolated value.

```
public virtual T Value { get; }
```

Property Value

T

The current value.

Methods

ForceSet(T)

Immediately sets the current and target values to a specified value.

```
public virtual void ForceSet(T newValue)
```

Parameters

newValue T

The value to set.

SetLerpSpeed(float)

Sets the speed at which the value interpolates towards the target.

```
public virtual void SetLerpSpeed(float newSpeed)
```

Parameters

newSpeed float

The new interpolation speed.

SetTarget(T)

Sets a new target value for interpolation.

```
public virtual void SetTarget(T newTarget)
```

Parameters

newTarget T

The new target value.

Update(float)

Updates the current value by interpolating towards the target value.

```
public virtual T Update(float deltaTime)
```

Parameters

deltaTime float

The time elapsed since the last update.

Returns

T

The updated current value after interpolation.

Class ObservableField<T>

Namespace: MGeLabs.Utils.[Data](#)

Assembly: cs.temp.dll.dll

Represents an observable field that notifies subscribers when its value changes.

```
public class ObservableField<T>
```

Type Parameters

T

Inheritance

object ← ObservableField<T>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Examples

The type of the value being observed.

```
ObservableField<int> observableInt = new ObservableField<int>(10);
observableInt.OnChange += newValue => Debug.Log($"Value changed to {newValue}");

observableInt.Value = 20; // Triggers OnChange and outputs: Value changed to 20
observableInt.Value = 20; // No event triggered because value is the same
```

Constructors

ObservableField(T)

Initializes a new instance of the [ObservableField<T>](#) class with an optional initial value.

```
public ObservableField(T initialValue = default)
```

Parameters

`initialValue` T

The initial value of the observable field.

Properties

Value

Gets or sets the value of the observable field.

```
public T Value { get; set; }
```

Property Value

T

Events

OnChange

Action to be invoked when the value changes.

```
public event Action<T> OnChange
```

Event Type

Action<T>

Class ObservableList<T>

Namespace: MGeLabs.Utils.[Data](#)

Assembly: cs.temp.dll.dll

Represents a list that notifies subscribers when items are added or removed.

```
public class ObservableList<T> : IEnumerable<T>
```

Type Parameters

T

The type of elements in the list.

Inheritance

object ← ObservableList<T>

Examples

```
ObservableList<string> observableList = new ObservableList<string>();
observableList.OnAdd += item => Debug.Log($"Added: {item}");
observableList.OnRemove += item => Debug.Log($"Removed: {item}");

observableList.Add("Hello"); // Output: Added: Hello
observableList.Add("World"); // Output: Added: World
observableList.Remove("Hello"); // Output: Removed: Hello
observableList.Clear(); // Output: Removed: World
```

Properties

Count

Gets the number of elements contained in the list.

```
public int Count { get; }
```

Property Value

int

this[int]

Gets the element at the specified index.

```
public T this[int index] { get; }
```

Parameters

index int

The zero-based index of the element to get.

Property Value

T

The element at the specified index.

Methods

Add(T)

Adds an item to the list and triggers the OnAdd event.

```
public void Add(T item)
```

Parameters

item T

The item to add to the list.

Clear()

Removes all items from the list and triggers the OnRemove event for each item.

```
public void Clear()
```

GetEnumerator()

Returns an enumerator that iterates through the list.

```
public IEnumerator<T> GetEnumerator()
```

Returns

`IEnumerator<T>`

An enumerator for the list.

Remove(T)

Removes the first occurrence of a specific item from the list and triggers the OnRemove event.

```
public bool Remove(T item)
```

Parameters

`item` T

The item to remove from the list.

Returns

`bool`

true if the item is successfully removed; otherwise, false.

Events

OnAdd

Event triggered when an item is added to the list.

```
public event Action<T> OnAdd
```

Event Type

Action<T>

OnRemove

Event triggered when an item is removed from the list.

```
public event Action<T> OnRemove
```

Event Type

Action<T>

Class WaitHelper

Namespace: MGeLabs.Utils.[Data](#)

Assembly: cs.temp.dll.dll

```
public static class WaitHelper
```

Inheritance

object ← WaitHelper

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

WaitForEndOfFrame

Gets a WaitForEndOfFrame instance.

```
public static WaitForEndOfFrame WaitForEndOfFrame { get; }
```

Property Value

WaitForEndOfFrame

Examples

```
// Usage example:  
IEnumerator ExampleCoroutine()  
{  
    yield return WaitHelper.WaitForEndOfFrame;  
}
```

WaitForFixedUpdate

Gets a WaitForFixedUpdate instance.

```
public static WaitForFixedUpdate WaitForFixedUpdate { get; }
```

Property Value

WaitForFixedUpdate

Examples

```
// Usage example:  
IEnumerator ExampleCoroutine()  
{  
    yield return WaitHelper.WaitForFixedUpdate;  
}
```

Methods

WaitForSeconds(float)

Gets a cached WaitForSeconds instance for the specified duration.

```
public static WaitForSeconds WaitForSeconds(float seconds)
```

Parameters

seconds float

The duration in seconds.

Returns

WaitForSeconds

A WaitForSeconds instance.

Examples


```
// Usage example:
IEnumerator ExampleCoroutine()
{
    // Wait for 1 second using cached instance
    yield return WaitHelper.WaitForSeconds(1f);
}
```

WaitForSecondsRealtime(float)

Gets a cached WaitForSecondsRealtime instance for the specified duration.

```
public static WaitForSecondsRealtime WaitForSecondsRealtime(float seconds)
```

Parameters

`seconds` float

The duration in seconds.

Returns

WaitForSecondsRealtime

A WaitForSecondsRealtime instance.

Examples

```
// Usage example:
IEnumerator ExampleCoroutine()
{
    // Wait for 1 second in real time using cached instance
    yield return WaitHelper.WaitForSecondsRealtime(1f);
}
```

Namespace MGeLabs.Utils.Debugging

Classes

[Logger](#)

A logger class that persists log entries and saves them to a file.

[NavMeshPathDrawer](#)

Draws the path of a NavMeshAgent using a LineRenderer.

Structs

[LogEntry](#)

Represents a log entry.

Enums

[ELogTypeMask](#)

Represents a bitmask for categorizing log types in debugging.

Enum ELogTypeMask

Namespace: MGeLabs.Utils.[Debugging](#)

Assembly: cs.temp.dll.dll

Represents a bitmask for categorizing log types in debugging.

```
public enum ELogTypeMask
```

Fields

All = -1

Assert = 8

Error = 4

Exception = 16

Log = 1

None = 0

Warning = 2

Struct LogEntry

Namespace: MGeLabs.Utils.[Debugging](#)

Assembly: cs.temp.dll.dll

Represents a log entry.

```
public struct LogEntry
```

Inherited Members

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

LogEntry(string, DateTime, string, string)

```
public LogEntry(string logType, DateTime time, string log, string stack)
```

Parameters

logType string

time DateTime

log string

stack string

Fields

dateTime

```
public DateTime dateTime
```

Field Value

DateTime

logString

```
public string logString
```

Field Value

string

stackTrace

```
public string stackTrace
```

Field Value

string

type

```
public string type
```

Field Value

string

Class Logger

Namespace: MGeLabs.Utils.[Debugging](#)

Assembly: cs.temp.dll.dll

A logger class that persists log entries and saves them to a file.

```
public class Logger : PersistentSingleton<Logger>
```

Inheritance

object ← Logger

Examples

Access the logger [instance](#) (singleton) [and](#) retrieve the current log as a string.
string currentLog = Logger.Instance.Log;

You can also customize ignored words [and](#) toggle stack trace appending in the inspector.

Properties

LogString

Gets the complete log as a string.

```
public string LogString { get; }
```

Property Value

string

Methods

Awake()

Initializes the logger.

```
protected override void Awake()
```

Events

OnNewLogEntry

```
public event Action<LogEntry> OnNewLogEntry
```

Event Type

Action<[LogEntry](#)>

Class NavMeshPathDrawer

Namespace: MGeLabs.Utils.[Debugging](#)

Assembly: cs.temp.dll.dll

Draws the path of a NavMeshAgent using a LineRenderer.

```
public class NavMeshPathDrawer : MonoBehaviour
```

Inheritance

object ← NavMeshPathDrawer

Examples

Attach this [component](#) to a GameObject [with](#) a NavMeshAgent [to](#) visualize the agent's path.

Example usage:

1. Add NavMeshAgent [component](#) to your GameObject.
2. Add this NavMeshPathDrawer [component](#).
3. Configure ['showPath'](#), ['pathColor'](#), and ['lineWidth'](#) in the Inspector.

The path will be drawn automatically every frame.

Namespace MGeLabs.Utils.Editor

Classes

[ColliderBoundsDrawer](#)

A MonoBehaviour that visualizes the bounds of a Collider in the Unity Editor.

Class ColliderBoundsDrawer

Namespace: MGeLabs.Utils.[Editor](#)

Assembly: cs.temp.dll.dll

A MonoBehaviour that visualizes the bounds of a Collider in the Unity Editor.

```
public class ColliderBoundsDrawer : MonoBehaviour
```

Inheritance

object ← ColliderBoundsDrawer

Namespace MGeLabs.Utls.EventSystem

Classes

[EventChannel<TChannel>](#)

Abstract class representing an event channel that can subscribe, unsubscribe, and publish events.

[EventManager](#)

Manages event channels and provides methods to subscribe and publish events.

Interfaces

[IEvent<TChannel>](#)

Interface for events with a specific event channel.

[IEventChannel](#)

Interface representing an event channel.

Class EventChannel<TChannel>

Namespace: MGeLabs.Utils.[EventSystem](#)

Assembly: cs.temp.dll.dll

Abstract class representing an event channel that can subscribe, unsubscribe, and publish events.

```
public abstract class EventChannel<TChannel> : IEventChannel where TChannel
: EventChannel<TChannel>
```

Type Parameters

TChannel

The type of the event channel.

Inheritance

object ← EventChannel<TChannel>

Implements

[IEventChannel](#)

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() ,
object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Examples

Define a custom event channel

```
[Preserve] // Add Preserve to prevent the compiler from stripping the class.
public class PlayerEventChannel : EventChannel<PlayerEventChannel>
{
}
```

Fields

handlers

Dictionary to store event handlers by event type and handler ID.

`protected readonly Dictionary<Type, Dictionary<int, Action<IEvent<TChannel>>>> handlers`

Field Value

Dictionary<Type, Dictionary<int, Action<[IEvent](#)<TChannel>>>>

handlersToRemove

Set of handler IDs to be removed after publishing.

`protected readonly HashSet<int> handlersToRemove`

Field Value

HashSet<int>

isPublishing

Indicates whether an event is currently being published.

`protected bool isPublishing`

Field Value

bool

needsClearAfterPublishing

Indicates whether the handlers dictionary needs to be cleared after publishing.

`protected bool needsClearAfterPublishing`

Field Value

bool

nextHandlerId

The next handler ID to be assigned.

```
protected int nextHandlerId
```

Field Value

int

Methods

HandleHandlersToRemove(Dictionary<int, Action<IEvent<TChannel>>>)

Handles the removal of handlers that were marked for removal during publishing.

```
protected virtual void HandleHandlersToRemove(Dictionary<int, Action<IEvent<TChannel>>> actions)
```

Parameters

actions Dictionary<int, Action<[IEvent](#)<TChannel>>>

The dictionary of actions to remove handlers from.

Publish<TEvent>(IEvent<TChannel>)

Publishes an event by invoking all handlers for the event type.

```
public virtual void Publish<TEvent>(IEvent<TChannel> @event) where TEvent : IEvent<TChannel>
```

Parameters

event [IEvent](#)<TChannel>

The event to publish.

Type Parameters

TEvent

The type of the event.

Subscribe<TEvent>(Action<TEvent>)

Subscribes a handler to an event type.

```
public virtual int Subscribe<TEvent>(Action<TEvent> handler) where TEvent : IEvent<TChannel>
```

Parameters

handler Action<TEvent>

The handler to be invoked when the event is published.

Returns

int

The ID of the subscribed handler.

Type Parameters

TEvent

The type of the event.

Unsubscribe<TEvent>(int)

Unsubscribes a handler from an event type.

```
public virtual void Unsubscribe<TEvent>(int handlerId) where TEvent : IEvent<TChannel>
```

Parameters

handlerId int

The ID of the handler to be unsubscribed.

Type Parameters

TEvent

The type of the event.

UnsubscribeAll()

Unsubscribes all event handlers.

```
public virtual void UnsubscribeAll()
```


Class EventManager

Namespace: MGeLabs.Utils.[EventSystem](#)

Assembly: cs.temp.dll.dll

Manages event channels and provides methods to subscribe and publish events.

```
public class EventManager : PersistentSingleton<EventManager>
```

Inheritance

object ← EventManager

Examples

```
// Publisher example
```

```
public class Player : MonoBehaviour
```

```
{
    public void Jump()
    {
        PlayerJumpEvent jumpEvent = new PlayerJumpEvent
        {
            jumpStrength = 5.0f
        };
        EventManager.Publish<PlayerEventChannel, PlayerJumpEvent>(jumpEvent);
    }
}
```

```
// Subscriber example
```

```
public class SoundManager : MonoBehaviour
```

```
{
    private int handle;

    private void OnEnable()
    {
        handle = EventManager.Subscribe<PlayerEventChannel, PlayerJumpEvent>(OnPlayerJump);
    }

    private void OnDisable()
    {
        EventManager.Unsubscribe<PlayerEventChannel, PlayerJumpEvent>(handle);
    }
}
```

```
private void OnPlayerJump(PlayerJumpEvent jumpEvent)
{
    // React to jump event, e.g. play sound
}
}
```

Fields

EventChannels

Dictionary to store event channels by their type.

```
protected static readonly Dictionary<Type, IEventChannel> EventChannels
```

Field Value

Dictionary<Type, [IEventChannel](#)>

Methods

Awake()

Called when the script instance is being loaded.

```
protected override void Awake()
```

GetEventChannel<T>()

Gets the event channel of the specified type.

```
protected static T GetEventChannel<T>() where T : EventChannel<T>
```

Returns

T

The event channel of the specified type.

Type Parameters

T

The type of the event channel.

Publish<TChannel, TEvent>(TEvent)

Publishes an event by invoking all handlers for the event type.

```
public static void Publish<TChannel, TEvent>(TEvent @event) where TChannel :  
    EventChannel<TChannel> where TEvent : IEvent<TChannel>
```

Parameters

event TEvent

The event to publish.

Type Parameters

TChannel

The type of the event channel.

TEvent

The type of the event.

RegisterEventChannels()

Registers all event channels by finding and instantiating classes that extend EventChannel.

```
protected static void RegisterEventChannels()
```

Subscribe<TChannel, TEvent>(Action<TEvent>)

Subscribes to an event by adding a handler to the event channel.

```
public static int Subscribe<TChannel, TEvent>(Action<TEvent> handler) where TChannel :  
EventChannel<TChannel> where TEvent : IEvent<TChannel>
```

Parameters

handler Action<TEvent>

The handler to add for the event.

Returns

int

The ID of the subscribed handler.

Type Parameters

TChannel

The type of the event channel.

TEvent

The type of the event.

Unsubscribe<TChannel, TEvent>(int)

Unsubscribes from an event by removing a handler from the event channel.

```
public static void Unsubscribe<TChannel, TEvent>(int handlerId) where TChannel :  
EventChannel<TChannel> where TEvent : IEvent<TChannel>
```

Parameters

handlerId int

The ID of the handler to remove for the event.

Type Parameters

TChannel

The type of the event channel.

TEvent

The type of the event.

UnsubscribeAll<TChannel>()

Unsubscribes all events by clearing the handlers dictionary in the specified event channel.

```
protected static void UnsubscribeAll<TChannel>() where TChannel : EventChannel<TChannel>
```

Type Parameters

TChannel

Interface IEvent<TChannel>

Namespace: MGeLabs.Utils.[EventSystem](#)

Assembly: cs.temp.dll.dll

Interface for events with a specific event channel.

```
public interface IEvent<TChannel> where TChannel : EventChannel<TChannel>
```

Type Parameters

TChannel

The type of the event channel.

Examples

Define an `event` 'PlayerJumpEvent' on the 'PlayerEventChannel' channel

```
public class PlayerJumpEvent : IEvent<PlayerEventChannel>
{
    public float jumpStrength;
}
```

Interface IEventChannel

Namespace: MGeLabs.Utils.[EventSystem](#)

Assembly: cs.temp.dll.dll

Interface representing an event channel.

```
public interface IEventChannel
```

Methods

UnsubscribeAll()

```
void UnsubscribeAll()
```

Namespace MGeLabs.Utils.Extensions

Classes

[AudioSourceExtension](#)

Provides extension methods for Audio sources.

[BoxColliderExtension](#)

Provides extension methods for the BoxCollider class.

[CameraExtension](#)

Provides extension methods for the Camera.

[ColorExtension](#)

Provides extension methods for the Color class.

[EnumExtension](#)

Provides extensions for enums.

[GameObjectExtension](#)

Provides extension methods for the GameObject class.

[LayerMaskExtension](#)

Provides extension methods for the LayerMask class.

[ListExtension](#)

Provides extension methods for lists.

[ObjectExtension](#)

Provides extension methods for objects to enhance logging functionality.

[QuaternionExtension](#)

Provides extension methods for the Quaternion class.

[RayExtension](#)

Provides extensions for Rays.

[RectExtension](#)

[RectTransformExtension](#)

Provides extension methods for RectTransforms.

[ScrollRectExtension](#)

Provides extension methods for ScrollRects.

[StringExtension](#)

Provides extension methods for Strings.

[TransformExtension](#)

Provides extension methods for the Transform class.

[Vector3Extension](#)

Provides extension methods for the Vector3 class.

Class AudioSourceExtension

Namespace: MGeLabs.Utils.[Extensions](#)

Assembly: cs.temp.dll.dll

Provides extension methods for Audio sources.

```
public static class AudioSourceExtension
```

Inheritance

object ← AudioSourceExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

FadeVolume(AudioSource, float, float, float?, Action)

Fades the volume of the given AudioSource from a start volume to a target volume over a specified duration.

```
public static IEnumerator FadeVolume(this AudioSource audioSource, float duration, float targetVolume, float? startVolume = null, Action onFinished = null)
```

Parameters

audioSource AudioSource

The AudioSource to fade.

duration float

The duration over which to fade the volume.

targetVolume float

The target volume to reach at the end of the fade.

startVolume float?

The starting volume. Defaults to 0.

onFinished Action

An optional callback to invoke when the fade is complete.

Returns

IEnumerator

An IEnumerator that can be used to run the fade operation in a coroutine.

Examples

```
// Usage example:
IEnumerator FadeExample(AudioSource audioSource)
{
    // Fade volume to 0 over 2 seconds
    yield return audioSource.FadeVolume(2f, 0f);

    // Fade back to full volume over 1.5 seconds
    yield return audioSource.FadeVolume(1.5f, 1f);
}
```

```
// Usage example:
StartCoroutine(audioSource.FadeVolume(1.5f, 1f));
```

PlayAndWaitUntilFinished(AudioSource, AudioClip, float, Action)

Plays an AudioClip on the given AudioSource and waits until it finishes playing, with an optional offset to end playback early and a callback when finished.

```
public static IEnumerator PlayAndWaitUntilFinished(this AudioSource audioSource, AudioClip
clip, float endTimeOffset = 0, Action onFinished = null)
```

Parameters

audioSource AudioSource

The AudioSource to play the clip on.

clip AudioClip

The AudioClip to play.

endTimeOffset float

The time (in seconds) to offset from the end of the clip. Defaults to 0. If the offset is greater than or equal to the clip length, the method exits early.

onFinished Action

An optional callback to invoke when the clip finishes playing.

Returns

IEnumerator

An IEnumerator that can be used to run the operation in a coroutine.

Examples

```
// Usage example:
IEnumerator PlayClip(AudioSource audioSource, AudioClip clip)
{
    yield return audioSource.PlayAndWaitUntilFinished(clip, 0.2f, () =>
        Debug.Log("Clip finished!"));
}
// StartCoroutine(PlayClip(audioSource, clip));
```

PlayOneShotWithVariance(AudioSource, AudioClip, float, float, float)

Plays an AudioClip with random pitch and volume variations for added audio diversity.

```
public static void PlayOneShotWithVariance(this AudioSource audioSource, AudioClip clip,
    float audioVolume = 1, float maxPitchVariation = 0.1, float maxVolumeVariation = 0.1)
```

Parameters

audioSource AudioSource

The AudioSource to play the clip on.

clip AudioClip

The AudioClip to play.

audioVolume float

The base volume at which to play the clip. Defaults to 1.

maxPitchVariation float

The maximum variation in pitch. Defaults to 0.1.

maxVolumeVariation float

The maximum variation in volume. Defaults to 0.1.

Examples

```
// Usage example:  
void PlayClipWithVariance(AudioSource audioSource, AudioClip clip)  
{  
    audioSource.PlayOneShotWithVariance(clip, 1f, 0.2f, 0.15f);  
}
```

TransitionToClip(AudioSource, AudioClip, float?, float, Action)

Transitions the AudioSource to a new AudioClip by fading out the current clip (if playing), stopping it, and then fading in the new clip.

```
public static IEnumerator TransitionToClip(this AudioSource audioSource, AudioClip clip,  
float? targetVolume = null, float transitionDuration = 1, Action onFinished = null)
```

Parameters

audioSource AudioSource

The AudioSource to transition.

`clip` AudioClip

The new AudioClip to play.

`targetVolume` float?

The target volume to fade to when playing the new clip. Defaults to the current volume of the AudioSource.

`transitionDuration` float

The total duration of the transition (fade out + fade in). Defaults to 1 second.

`onFinished` Action

An optional callback to invoke when the transition is finished.

Returns

IEnumerator

An IEnumerator that can be used to run the transition in a coroutine.

Examples

```
// Usage example:
IEnumerator TransitionExample(AudioSource audioSource, AudioClip newClip)
{
    yield return audioSource.TransitionToClip(newClip, 0.8f, 2f);
}
// StartCoroutine(TransitionExample(audioSource, newClip));
```

WaitUntilFinished(AudioSource, float, Action)

Waits until the current clip on the AudioSource finishes playing, with an optional offset to end early and a callback when finished.

```
public static IEnumerator WaitUntilFinished(this AudioSource audioSource, float
endTimeOffset = 0, Action onFinished = null)
```

Parameters

audioSource AudioSource

The AudioSource playing the clip.

endTimeOffset float

The time (in seconds) to offset from the end of the clip. Defaults to 0. If the offset is greater than or equal to the clip length, the method exits early.

onFinished Action

An optional callback to invoke when the clip finishes playing.

Returns

IEnumerator

An IEnumerator to be used in a coroutine.

Examples

```
yield return audioSource.WaitUntilFinished(0.1f, () => Debug.Log("Done"));
```

Class BoxColliderExtension

Namespace: MGeLabs.Utils.[Extensions](#)

Assembly: cs.temp.dll.dll

Provides extension methods for the BoxCollider class.

```
public static class BoxColliderExtension
```

Inheritance

object ← BoxColliderExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

RandomPoint(BoxCollider, bool)

Returns a random point within the bounds of the BoxCollider.

```
public static Vector3 RandomPoint(this BoxCollider collider, bool includeY = false)
```

Parameters

collider BoxCollider

includeY bool

If true, includes the Y coordinate in the random point; otherwise, uses the Y position of the collider's transform.

Returns

Vector3

A random point within the bounds of the BoxCollider.

Examples

```
// Usage example:
void Example(BoxCollider boxCollider)
{
    // Get a random point inside the collider ignoring Y (using collider's Y position)
    Vector3 point = boxCollider.RandomPoint();

    // Get a random point inside the collider including Y coordinate
    Vector3 pointWithY = boxCollider.RandomPoint(true);

    Debug.Log("Random point (no Y): " + point);
    Debug.Log("Random point (with Y): " + pointWithY);
}
```

Class CameraExtension

Namespace: MGeLabs.Utils.[Extensions](#)

Assembly: cs.temp.dll.dll

Provides extension methods for the Camera.

```
public static class CameraExtension
```

Inheritance

object ← CameraExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Capture(Camera, Action<byte[]>, LayerMask)

Captures the current camera view and invokes a callback with the image data as a byte array.

```
public static void Capture(this Camera camera, Action<byte[]> callback, LayerMask  
excludedLayers = null)
```

Parameters

camera Camera

The Camera instance to capture the view from.

callback Action<byte[]>

The callback to invoke with the captured image data.

excludedLayers LayerMask

The layers to exclude from the camera capture. Defaults to none.

Examples

```
// Usage example:
void Start()
{
    Camera mainCam = Camera.main;
    mainCam.Capture(OnCaptured);
}

void OnCaptured(byte[] imageData)
{
    Debug.Log("Captured image byte size: " + imageData.Length);
    // You can now save the image or use it however you want
}
```

Raycast(Camera, out RaycastHit, LayerMask, Vector3?, float, QueryTriggerInteraction, bool)

Performs a raycast from the camera's screen point and returns whether a hit occurred.

```
public static bool Raycast(this Camera camera, out RaycastHit hit, LayerMask includedLayers,
    Vector3? position = null, float maxDistance = 3.4028235E+38, QueryTriggerInteraction
    queryTriggerInteraction = null, bool drawRay = false)
```

Parameters

camera Camera

The Camera instance to perform the raycast from.

hit RaycastHit

The RaycastHit object containing information about the hit.

includedLayers LayerMask

The layers to include in the raycast.

position Vector3?

The screen position to cast the ray from. Defaults to the center of the screen.

maxDistance float

The maximum distance for the raycast. Defaults to infinity.

queryTriggerInteraction QueryTriggerInteraction

Specifies whether to include trigger colliders in the raycast.

drawRay bool

Indicates whether to draw the ray in the scene view for debugging purposes.

Returns

bool

true if the raycast hit an object; otherwise, **false**.

Examples

Example usage:

```
RaycastHit hit;
bool didHit = Camera.main.Raycast(
    includedLayers: LayerMask.GetMask("Default"),
    hit: out hit,
    drawRay: true
);

if (didHit)
{
    Debug.Log($"Hit: {hit.collider.name}");
}
```

Class ColorExtension

Namespace: MGeLabs.Utils.[Extensions](#)

Assembly: cs.temp.dll.dll

Provides extension methods for the Color class.

```
public static class ColorExtension
```

Inheritance

object ← ColorExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

AdjustSaturation(Color, float)

Adjusts the saturation and brightness of the given color by the specified intensity.

```
public static Color AdjustSaturation(this Color color, float intensity)
```

Parameters

color Color

The original color to adjust.

intensity float

The intensity by which to adjust the saturation and value. Positive values increase, negative values decrease.

Returns

Color

A new color with adjusted saturation and value.

Examples

```
Color original = Color.red;
Color adjusted = original.AdjustSaturation(0.2f);
Debug.Log("Adjusted color: " + adjusted);
```

FromHex(string)

Converts a hexadecimal string to a Color.

```
public static Color FromHex(this string hex)
```

Parameters

hex string

The hexadecimal string to convert.

Returns

Color

The Color represented by the hexadecimal string.

Examples

```
string hex = "#FF00FF";
try
{
    Color color = hex.FromHex();
    Debug.Log("Color: " + color);
}
catch (Exception e)
{
    Debug.LogError(e.Message);
}
```

Exceptions

System.ArgumentException

System.ArgumentException: Thrown when the hex string is in an invalid format.

ToHex(Color)

Converts the color to a hexadecimal string.

```
public static string ToHex(this Color color)
```

Parameters

color Color

The color to convert.

Returns

string

A hexadecimal string representation of the color.

Examples

```
Color color = Color.green;  
string hex = color.ToHex();  
Debug.Log("Hex: " + hex); // Output: #00FF00
```

WithAlpha(Color, float)

Creates a new color with the same RGB values but a modified alpha value.

```
public static Color WithAlpha(this Color color, float alpha)
```

Parameters

color Color

The original color.

alpha float

The new alpha value.

Returns

Color

A new color with the specified alpha value.

Examples

```
Color original = Color.blue;  
Color withAlpha = original.WithAlpha(0.5f);  
Debug.Log("Color with new alpha: " + withAlpha);
```


Class EnumExtension

Namespace: MGeLabs.Utils.[Extensions](#)

Assembly: cs.temp.dll.dll

Provides extensions for enums.

```
public static class EnumExtension
```

Inheritance

object ← EnumExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

RandomValue<T>()

Returns a random value from the specified enum type.

```
public static T RandomValue<T>() where T : Enum
```

Returns

T

A random value from the enum.

Type Parameters

T

The enum type.

Examples

```
// Example enum
enum Colors { Red, Green, Blue }

// Get a random color
Colors randomColor = EnumExtension.RandomValue<Colors>();
Debug.Log(randomColor);
```

Class GameObjectExtension

Namespace: MGeLabs.Utils.[Extensions](#)

Assembly: cs.temp.dll.dll

Provides extension methods for the GameObject class.

```
public static class GameObjectExtension
```

Inheritance

object ← GameObjectExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetOrAddComponent<T>(GameObject)

Retrieves a component of the specified type from the GameObject. If the component does not exist, it adds a new one to the GameObject.

```
public static T GetOrAddComponent<T>(this GameObject obj) where T : Component
```

Parameters

obj GameObject

The GameObject to retrieve or add the component to.

Returns

T

The existing or newly added component of the specified type.

Type Parameters

T

The type of the component to retrieve or add.

Examples

```
// Usage example:  
GameObject obj = new GameObject("MyObject");  
Rigidbody rb = obj.GetOrAddComponent<Rigidbody>();
```

SetLayersRecursively(GameObject, LayerMask, bool)

Recursively sets the of this GameObject and all its children, returning a dictionary of each GameObject to its original layer mask.

```
public static Dictionary<GameObject, LayerMask> SetLayersRecursively(this GameObject obj,  
LayerMask layerMask, bool includeInactive = true)
```

Parameters

obj GameObject

The root GameObject whose layer (and its descendants) will be set.

layerMask LayerMask

The target layer mask. Internally, is an int index, so this method uses **layerMask.value** as the new layer index.

includeInactive bool

If **true**, will include inactive children; otherwise only active ones.

Returns

Dictionary<GameObject, LayerMask>

A System.Collections.Generic.Dictionary`2 mapping each affected GameObject to its original (as a).

Class LayerMaskExtension

Namespace: MGeLabs.Utils.[Extensions](#)

Assembly: cs.temp.dll.dll

Provides extension methods for the LayerMask class.

```
public static class LayerMaskExtension
```

Inheritance

object ← LayerMaskExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

CreateLayerMaskFromIndices(params int[])

Creates a LayerMask from an array of layer indices.

```
public static LayerMask CreateLayerMaskFromIndices(params int[] layerIndices)
```

Parameters

layerIndices int[]

An array of layer indices.

Returns

LayerMask

A LayerMask representing the specified layers.

Examples

```
// Create a LayerMask for layers 0, 2, and 5
LayerMask mask = LayerMaskExtension.CreateLayerMaskFromIndices(0, 2, 5);
```

Exceptions

System.ArgumentException

Thrown when a layer index is out of the valid range (0-31).

CreateLayerMaskFromNames(params string[])

Creates a LayerMask from an array of layer names.

```
public static LayerMask CreateLayerMaskFromNames(params string[] layerNames)
```

Parameters

layerNames string[]

An array of layer names to convert into a LayerMask.

Returns

LayerMask

A LayerMask representing the specified layers.

Examples

```
// Create a LayerMask from layer names
LayerMask mask = LayerMaskExtension.NamesToLayer("Default", "UI", "Player");
```

GetSingleLayerIndex(LayerMask)

Retrieves the index of a single layer from the LayerMask.

```
public static int GetSingleLayerIndex(this LayerMask mask)
```

Parameters

mask LayerMask

The LayerMask to extract the layer index from.

Returns

int

The index of the single layer in the LayerMask. Returns 0 and logs an error if the LayerMask contains no layers or multiple layers.

Examples

```
LayerMask mask = LayerMaskExtension.CreateLayerMaskFromIndices(8);  
int index = mask.GetSingleLayerIndex(); // index will be 8
```

Class ListExtension

Namespace: MGeLabs.Utils.[Extensions](#)

Assembly: cs.temp.dll.dll

Provides extension methods for lists.

```
public static class ListExtension
```

Inheritance

object ← ListExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Center(List<Vector3>)

Calculates the center point of a list of Vector3s.

```
public static Vector3 Center(this List<Vector3> list)
```

Parameters

list List<Vector3>

The list of Vector3s to calculate the center point of.

Returns

Vector3

The center point of the list of Vector3s.

Examples


```
List<Vector3> positions = new()  
{  
    new Vector3(1, 0, 0),  
    new Vector3(0, 1, 0),  
    new Vector3(0, 0, 1)  
};  
Vector3 centerPoint = positions.Center();  
// centerPoint is approximately (0.33, 0.33, 0.33)
```

Chunk<T>(List<T>, int)

Splits a list into smaller chunks of a specified size.

```
public static List<List<T>> Chunk<T>(this List<T> list, int chunkSize)
```

Parameters

list List<T>

The source list to split into chunks.

chunkSize int

The size of each chunk.

Returns

List<List<T>>

A list of smaller lists, each containing up to **chunkSize** items.

Type Parameters

T

The type of items in the list.

Clone<T>(List<T>)

Creates a shallow copy of the list.

```
public static List<T> Clone<T>(this List<T> list)
```

Parameters

list List<T>

The list to clone.

Returns

List<T>

A new list containing the same elements as the original list.

Type Parameters

T

The type of elements in the list.

Examples

```
List<float> original = new() { 1.0f, 2.0f, 3.0f };  
List<float> copy = original.Clone();  
// copy contains the same elements as original
```

Flatten<T>(List<List<T>>)

Flattens a list of lists into a single list by concatenating all inner lists.

```
public static List<T> Flatten<T>(this List<List<T>> listOfLists)
```

Parameters

listOfLists List<List<T>>

The list of lists to flatten.

Returns

List<T>

A single list containing all elements from the inner lists.

Type Parameters

T

The type of elements in the lists.

Examples

```
List<List<int>> nested = new()
{
    new List<int> { 1, 2 },
    new List<int> { 3, 4 },
    new List<int> { 5 }
};
List<int> flat = nested.Flatten();
// flat is { 1, 2, 3, 4, 5 }
```

Furthest(List<Transform>, Transform)

Finds the furthest transform in the list from the specified transform.

```
public static Transform Furthest(this List<Transform> transformList,
    Transform compareTransform)
```

Parameters

transformList List<Transform>

The list of transforms to search.

compareTransform Transform

The transform to compare against.

Returns

Transform

The furthest transform from the specified transform.

Examples

```
List<Transform> enemies = new List<Transform>();
Transform player = someGameObject.transform;
Transform furthestEnemy = enemies.Furthest(player);
// furthestEnemy is the furthest enemy transform from the player
```

Lerp(List<float>, float)

Linearly interpolates between elements in a list of floats based on a parameter t.

```
public static float Lerp(this List<float> list, float t)
```

Parameters

list List<float>

The list of floats to interpolate.

t float

The interpolation parameter, typically between 0 and 1.

Returns

float

The interpolated float value.

Examples

```
List<float> points = new() { 0f, 10f, 20f };
float value = points.Lerp(0.25f);
// value is interpolated between 0 and 10, roughly 2.5
```

Exceptions

System.ArgumentException

System.ArgumentException: Thrown when the list is empty.

Nearest(List<Transform>, Transform)

Finds the nearest transform in the list to the specified transform.

```
public static Transform Nearest(this List<Transform> transformList,  
    Transform compareTransform)
```

Parameters

transformList List<Transform>

The list of transforms to search.

compareTransform Transform

The transform to compare against.

Returns

Transform

The nearest transform to the specified transform.

Examples

```
List<Transform> enemies = new List<Transform>();  
Transform player = someGameObject.transform;  
Transform nearestEnemy = enemies.Nearest(player);  
// nearestEnemy is the closest enemy transform to the player
```

Next<T>(List<T>, T)

Returns the next element in the list after the specified item. If the item is not found, returns the first element if the list is not empty, otherwise returns the default value for the type.

```
public static T Next<T>(this List<T> list, T item)
```

Parameters

list List<T>

The list to search.

item T

The item to find the next element of.

Returns

T

The next element in the list after the specified item, or the first element if the item is not found and the list is not empty, otherwise the default value for the type.

Type Parameters

T

The type of elements in the list.

Examples

```
List<string> colors = new() { "Red", "Green", "Blue" };
string nextColor = colors.Next("Green");
// nextColor is "Blue"
string nextOfUnknown = colors.Next("Yellow");
// nextOfUnknown is "Red" (first element) because "Yellow" not found
```

ProcessInBatches<T>(List<T>, Action<T>, int, float, Action<T>, Action<T, bool>, Action<IReadOnlyList<T>>, Action<IReadOnlyList<T>>, Action<IReadOnlyList<T>>, Action<T, Exception>)

Executes a batch operation on a list of items, processing them in groups with optional delays and callbacks.

```
public static IEnumerable ProcessInBatches<T>(this List<T> itemsList, Action<T>
actionOnItem, int itemsPerBatch = 5, float batchDelay = 0.1, Action<T> onItemProcessStart =
```

```
    null, Action<T, bool> onItemProcessFinished = null, Action<IReadOnlyList<T>> onBatchStart =  
    null, Action<IReadOnlyList<T>> onBatchFinished = null, Action<IReadOnlyList<T>> onFinished =  
    null, Action<T, Exception> onError = null)
```

Parameters

itemsList List<T>

The list of items to process in batches.

actionOnItem Action<T>

The action to execute on each item in the batch.

itemsPerBatch int

The number of items to process in each batch. Default is 5.

batchDelay float

The delay (in seconds) between processing each batch. Default is 0.1f.

onItemProcessStart Action<T>

Optional callback invoked before processing each item; receives the item as **T**.

onItemProcessFinished Action<T, bool>

Optional callback invoked after processing each item; receives the item as **T** and a **bool** indicating success.

onBatchStart Action<IReadOnlyList<T>>

Optional callback invoked at the start of each batch; receives the current batch as a **IReadOnlyList<T>**.

onBatchFinished Action<IReadOnlyList<T>>

Optional callback invoked at the end of each batch; receives the processed batch as a **IReadOnlyList<T>**.

onFinished Action<IReadOnlyList<T>>

Optional callback invoked after all batches are processed; receives the entire list as a **IReadOnlyList<T>**.

onError Action<T, Exception>

Optional callback invoked when an error occurs during item processing; receives the item that failed (T) and the thrown `Exception`.

Returns

IEnumerator

An enumerator that can be used to execute the batch operation.

Type Parameters

T

The type of items in the list.

Examples

```
List<int> numbers = new() { 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 };
IEnumerator batch = numbers.ProcessInBatches(
    item => Debug.Log($"Processing: {item}"),
    itemsPerBatch: 3,
    batchDelay: 0.5f,
    onBatchStart: batchItems => Debug.Log($"Batch started: {string.Join(", ", batchItems)}"),
    onBatchFinished: batchItems => Debug.Log($"Batch finished: {string.Join(", ", batchItems)}"),
    onFinished: allItems => Debug.Log("All batches complete!")
);
// Use StartCoroutine(batch) in a MonoBehaviour to execute.
```

ProcessInBatches<T>(List<T>, Func<T, IEnumerator>, int, float, Action<T>, Action<T, bool>, Action<IReadOnlyList<T>>, Action<IReadOnlyList<T>>, Action<IReadOnlyList<T>>)

Processes a list of items in batches, where each item is processed using a coroutine (IEnumerator).

```
public static IEnumerator ProcessInBatches<T>(this List<T> itemsList, Func<T, IEnumerator>
routineOnItem, int itemsPerBatch = 5, float batchDelay = 0.1, Action<T> onItemProcessStart
= null, Action<T, bool> onItemProcessFinished = null, Action<IReadOnlyList<T>> onBatchStart
= null, Action<IReadOnlyList<T>> onBatchFinished = null, Action<IReadOnlyList<T>> onFinished
= null)
```


Parameters

itemsList List<T>

The list of items to process in batches.

routineOnItem Func<T, IEnumerator>

The coroutine to execute for each item.

itemsPerBatch int

The number of items to process per batch. Defaults to 5.

batchDelay float

The delay (in seconds) between processing batches. Defaults to 0.1f.

onItemProcessStart Action<T>

Optional callback invoked before processing each item; receives the item as **T**.

onItemProcessFinished Action<T, bool>

Optional callback invoked after processing each item; receives the item as **T** and a **bool** indicating success.

onBatchStart Action<IReadOnlyList<T>>

Optional callback invoked before processing each batch; receives the batch as **IReadOnlyList<T>**.

onBatchFinished Action<IReadOnlyList<T>>

Optional callback invoked when a batch is completed; receives the batch as **IReadOnlyList<T>**.

onFinished Action<IReadOnlyList<T>>

Optional callback invoked when all items have been processed; receives the full list of processed items as **IReadOnlyList<T>**.

Returns

IEnumerator

An enumerator that can be used to execute the batches over time.

Type Parameters

T

The type of items in the list.

Random<T>(List<T>, int)

Returns a list of unique random elements from the list.

```
public static List<T> Random<T>(this List<T> list, int count)
```

Parameters

list List<T>

The list to select random elements from.

count int

The number of unique random elements to return.

Returns

List<T>

A list containing unique random elements from the original list.

Type Parameters

T

The type of elements in the list.

Examples

```
List<int> numbers = new() { 1, 2, 3, 4, 5 };  
List<int> randomSubset = numbers.Random(3);  
// randomSubset contains 3 unique elements randomly selected from numbers
```

Exceptions

System.ArgumentException

Thrown if count is greater than the list count or if count is negative.

Random<T>(List<T>)

Returns a random element from the list.

```
public static T Random<T>(this List<T> list)
```

Parameters

list List<T>

The list to select a random element from.

Returns

T

A random element from the list.

Type Parameters

T

The type of elements in the list.

Examples

```
List<int> values = new() { 10, 20, 30 };  
int randomValue = values.Random();  
// randomValue is one of 10, 20, or 30
```

Exceptions

System.ArgumentException

System.ArgumentException: Thrown when the list is empty.

RandomWithProbability<T>(List<T>, List<float>)

Returns one random element from `list` using weighted probabilities. Each value in `probabilities` corresponds by index to an item in `list`.

```
public static T RandomWithProbability<T>(this List<T> list, List<float> probabilities)
```

Parameters

`list` List<T>

Source items to choose from.

`probabilities` List<float>

Weight values for each item. Must have the same number of elements as `list`. Higher values increase the chance of selecting the corresponding item.

Returns

T

A randomly selected item from `list` based on the provided weights.

Type Parameters

T

The type of element in the list.

Examples

```
List<string> loot = new() { "Common", "Rare", "Epic" };
List<float> weights = new() { 70f, 25f, 5f };

string result = loot.RandomWithProbability(weights);
// result is most likely "Common", less likely "Rare", and rarely "Epic"
```

Exceptions

System.ArgumentException

Thrown when `list` is empty, when list sizes do not match, or when the sum of all probabilities is less than or equal to zero.

Replace<T>(List<T>, T, T)

Replaces the first occurrence of a specified item in the list with a new item.

```
public static List<T> Replace<T>(this List<T> list, T oldItem, T newItem)
```

Parameters

`list` List<T>

The list in which the replacement will occur.

`oldItem` T

The item to be replaced.

`newItem` T

The item to replace the old item with.

Returns

List<T>

The original list with the specified item replaced.

Type Parameters

`T`

The type of elements in the list.

Examples

```
List<string> fruits = new() { "Apple", "Banana", "Cherry" };  
fruits.Replace("Banana", "Orange");  
// fruits is now { "Apple", "Orange", "Cherry" }
```

Shuffle<T>(List<T>)

Shuffles the elements of a list in place using the Fisher-Yates algorithm.

```
public static List<T> Shuffle<T>(this List<T> list)
```

Parameters

list List<T>

The list to shuffle.

Returns

List<T>

Type Parameters

T

The type of elements in the list.

Examples

```
List<string> names = new() { "Alice", "Bob", "Charlie" };  
names.Shuffle();  
// names order is now randomized, e.g., "Charlie", "Alice", "Bob"
```

SortInPlace<T>(List<T>, Comparison<T>)

Sorts the elements of the list in place using the specified comparison.

```
public static List<T> SortInPlace<T>(this List<T> list, Comparison<T> comparison)
```

Parameters

list List<T>

The list to sort.

comparison Comparison<T>

The comparison to use for sorting the elements.

Returns

List<T>

The sorted list.

Type Parameters

T

The type of elements in the list.

Examples

```
List<int> numbers = new() { 5, 3, 8, 1 };  
numbers.SortInPlace((a, b) => a.CompareTo(b));  
// numbers is now sorted: 1, 3, 5, 8
```

Class ObjectExtension

Namespace: MGeLabs.Utils.[Extensions](#)

Assembly: cs.temp.dll.dll

Provides extension methods for objects to enhance logging functionality.

```
public static class ObjectExtension
```

Inheritance

object ← ObjectExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Log(object, object)

Logs a message with the object's type prefix.

```
public static void Log(this object obj, object message)
```

Parameters

obj object

The object to log the message for.

message object

The message to log.

Examples

```
this.Log("Hello, World!");
```


Log(object, params object[])

Logs multiple messages with the object's type prefix.

```
public static void Log(this object obj, params object[] messages)
```

Parameters

obj object

The object to log the messages for.

messages object[]

The messages to log.

Examples

```
this.Log("Health:", playerHealth, "Score:", playerScore);
```

LogError(object, object)

Logs an error message with the object's type prefix.

```
public static void LogError(this object obj, object message)
```

Parameters

obj object

The object to log the error message for.

message object

The error message to log.

Examples

```
this.LogError("Player died unexpectedly.");
```

LogError(object, params object[])

Logs multiple error messages with the object's type prefix.

```
public static void LogError(this object obj, params object[] messages)
```

Parameters

obj object

The object to log the error messages for.

messages object[]

The error messages to log.

Examples

```
this.LogError("Error code:", errorCode, "Details:", errorDetails);
```

LogException(object, Exception)

Logs an exception with the object's type prefix.

```
public static void LogException(this object obj, Exception exception)
```

Parameters

obj object

The object to log the exception for.

exception Exception

The exception to log.

Examples

```
try  
{
```

```
    // some code that may throw
}
catch (Exception ex)
{
    this.LogException(ex);
}
```

LogException(object, string, Exception)

Logs an exception with a custom message and the object's type prefix.

```
public static void LogException(this object obj, string message, Exception exception)
```

Parameters

obj object

The object to log the exception for.

message string

The custom message to include with the exception.

exception Exception

The exception to log.

Examples

```
try
{
    // some code that may throw
}
catch (Exception ex)
{
    this.LogException("Custom error message", ex);
}
```

LogWarning(object, object)

Logs a warning message with the object's type prefix.

```
public static void LogWarning(this object obj, object message)
```

Parameters

obj object

The object to log the warning message for.

message object

The warning message to log.

Examples

```
this.LogWarning("Low health detected!");
```

LogWarning(object, params object[])

Logs multiple warning messages with the object's type prefix.

```
public static void LogWarning(this object obj, params object[] messages)
```

Parameters

obj object

The object to log the warning messages for.

messages object[]

The warning messages to log.

Examples

```
this.LogWarning("Low health:", health, "Ammo:", ammoCount);
```

Class QuaternionExtension

Namespace: MGeLabs.Utils.[Extensions](#)

Assembly: cs.temp.dll.dll

Provides extension methods for the Quaternion class.

```
public static class QuaternionExtension
```

Inheritance

object ← QuaternionExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

With(Quaternion, float?, float?, float?)

Returns a new Quaternion with the specified Euler angles applied.

```
public static Quaternion With(this Quaternion quaternion, float? x = null, float? y = null, float? z = null)
```

Parameters

quaternion Quaternion

The original Quaternion.

x float?

The x component of the Euler angles to apply. Defaults to 0 if null.

y float?

The y component of the Euler angles to apply. Defaults to 0 if null.

z float?

The z component of the Euler angles to apply. Defaults to 0 if null.

Returns

Quaternion

A new Quaternion with the specified Euler angles applied.

Examples

```
Quaternion original = Quaternion.Euler(10, 20, 30);  
Quaternion modified = original.With(y: 45);  
Debug.Log(modified.eulerAngles); // Output: (10, 45, 30)
```

Class RayExtension

Namespace: MGeLabs.Utils.[Extensions](#)

Assembly: cs.temp.dll.dll

Provides extensions for Rays.

```
public static class RayExtension
```

Inheritance

object ← RayExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Draw(Ray, Color, float, float)

Draws a ray in the Unity editor for debugging purposes.

```
public static void Draw(this Ray ray, Color color, float length = 100, float duration = 0.25)
```

Parameters

ray Ray

The ray to draw.

color Color

The color of the ray.

length float

The length of the ray (default is 100f).

duration float

The duration the ray will be visible in seconds (default is 0.25f).

Examples

```
Ray ray = new Ray(transform.position, transform.forward);  
ray.Draw(Color.red, 50f, 1f);
```


Class RectExtension

Namespace: MGeLabs.Utils.[Extensions](#)

Assembly: cs.temp.dll.dll

```
public static class RectExtension
```

Inheritance

object ← RectExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ToLocalSpace(Rect, Transform)

Converts a Rect from world space to local space.

```
public static Rect ToLocalSpace(this Rect r, Transform transform)
```

Parameters

r Rect

The Rect to convert.

transform Transform

The Transform to use for conversion.

Returns

Rect

A Rect in local space.

Examples

```
Rect worldRect = new Rect(0, 0, 1, 1);  
Rect localRect = worldRect.ToLocalSpace(transform);
```

ToWorldSpace(Rect, Transform)

Converts a Rect from local space to world space.

```
public static Rect ToWorldSpace(this Rect r, Transform transform)
```

Parameters

r Rect

The Rect to convert.

transform Transform

The Transform to use for conversion.

Returns

Rect

A Rect in world space.

Examples

```
Rect localRect = new Rect(0, 0, 1, 1);  
Rect worldRect = localRect.ToWorldSpace(transform);
```

Class RectTransformExtension

Namespace: MGeLabs.Utils.[Extensions](#)

Assembly: cs.temp.dll.dll

Provides extension methods for RectTransforms.

```
public static class RectTransformExtension
```

Inheritance

object ← RectTransformExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

RebuildLayout(RectTransform)

Rebuilds the layout of the rect transform to fix potential sizing issues and UI glitches.

```
public static IEnumerator RebuildLayout(this RectTransform rect)
```

Parameters

rect RectTransform

The RectTransform of object.

Returns

IEnumerator

An IEnumerator for the coroutine.

Examples

```
StartCoroutine(myRectTransform.RebuildLayout());
```

Class ScrollRectExtension

Namespace: MGeLabs.Utils.[Extensions](#)

Assembly: cs.temp.dll.dll

Provides extension methods for ScrollRects.

```
public static class ScrollRectExtension
```

Inheritance

object ← ScrollRectExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ResetContentScroll(ScrollRect, float, int)

Resets the scroll position of a ScrollRect to the specified value (default is 1, which corresponds to the left/top).

```
public static IEnumerator ResetContentScroll(this ScrollRect scrollRect, float scroll = 1, int direction = -1)
```

Parameters

scrollRect ScrollRect

The ScrollRect component whose scroll position will be reset.

scroll float

The normalized scroll position to set (0 = right/bottom, 1 = left/top). Default is 1.

direction int

The scroll direction to reset: 0 = vertical, 1 = horizontal, -1 = both. Default is -1 (both).

Returns

IEnumerator

- | An IEnumerator to be used as a coroutine.

Class StringExtension

Namespace: MGeLabs.Utils.[Extensions](#)

Assembly: cs.temp.dll.dll

Provides extension methods for Strings.

```
public static class StringExtension
```

Inheritance

object ← StringExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Sanitize(string, string)

Sanitizes the input string by replacing spaces with underscores and removing special characters.

```
public static string Sanitize(this string text, string allowedPattern = "A-Za-z0-9_")
```

Parameters

text string

The input string to be sanitized.

allowedPattern string

A regex pattern defining the set of allowed characters. Defaults to "A-Za-z0-9_", which allows alphanumeric characters and underscores.

Returns

string

A sanitized string with spaces replaced by underscores and disallowed characters removed.

Examples

```
string sanitized = "Hello, World!".Sanitize();  
// sanitized == "Hello_World"
```

Swap(string, string, string)

Swaps occurrences of two specified substrings within the given text.

```
public static string Swap(this string text, string swapA, string swapB)
```

Parameters

text string

The original text where the swap will occur.

swapA string

The first substring to swap.

swapB string

The second substring to swap.

Returns

string

A new string with the specified substrings swapped.

Examples

```
string result = "hello world".Swap("hello", "world");  
// result == "world hello"
```

Truncate(string, int, string)

Truncates the input string to a specified maximum length without cutting off words. Appends a truncation indicator if necessary.

```
public static string Truncate(this string text, int maxLength, string indicator = "...")
```

Parameters

text string

The input string to be truncated.

maxLength int

The maximum length of the truncated string (not counting the indicator).

indicator string

The string to append if truncation occurs. Defaults to "...".

Returns

string

A truncated string that ends on a word boundary (space/tab/newline) with the truncation indicator appended. Returns the original string if it's shorter than or equal to `maxLength`. Returns an empty string if the input is null or `maxLength` is negative.

Examples

```
string t1 = "This is a long string".Truncate(10);
```

```
// t1 == "This is a..."
```

```
string t2 = "Short".Truncate(10);
```

```
// t2 == "Short"
```

```
string t3 = null.Truncate(5);
```

```
// t3 == ""
```

```
string t4 = "Supercalifragilistic".Truncate(5);
```

```
// t4 == "Super..."
```

Class TransformExtension

Namespace: MGeLabs.Utils.[Extensions](#)

Assembly: cs.temp.dll.dll

Provides extension methods for the Transform class.

```
public static class TransformExtension
```

Inheritance

object ← TransformExtension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

DestroyChildren(Transform)

Destroys all child GameObjects of the given transform.

```
public static void DestroyChildren(this Transform transform)
```

Parameters

transform Transform

The transform whose children will be destroyed.

FindInHierarchy(Transform, Func<Transform, bool>, bool)

Finds all Transforms in the hierarchy of the given root Transform that match the specified filter.

```
public static Transform[] FindInHierarchy(this Transform rootTransform, Func<Transform, bool> filter, bool includeInactive = false)
```

Parameters

rootTransform Transform

The root Transform to search within.

filter Func<Transform, bool>

A function that defines the condition each Transform must satisfy to be included in the results.

includeInactive bool

If true, includes inactive Transforms in the search.

Returns

Transform[]

An array of Transforms that match the filter condition. Returns an empty array if no matches are found.

Examples

```
// Find all transforms whose name contains "Hand".
Transform[] hands = rootTransform.FindInHierarchy(
    t => t.name.Contains("Hand")
);
```

MoveAndRotateToTransform(Transform, Transform, float, bool, Action)

Smoothly moves a Transform to match the position and rotation of a target Transform over a specified duration. Can optionally move in local space.

```
public static IEnumerator MoveAndRotateToTransform(this Transform target, Transform
destination, float duration = 0.5, bool useLocal = true, Action onFinished = null)
```

Parameters

target Transform

The Transform to move.

destination Transform

The target Transform to move towards.

duration float

The duration of the movement in seconds. Defaults to 0.5 seconds.

useLocal bool

If true, operates on localPosition/localRotation instead of world position/rotation.

onFinished Action

An optional callback to invoke when the rotation is complete.

Returns

IEnumerator

An IEnumerator to be used in a coroutine.

Examples

```
StartCoroutine(characterTransform.MoveAndRotateToTransform(spawnTransform, 1.2f));
```

MoveToPosition(Transform, Vector3, float, bool, Action)

Smoothly moves a Transform to a specified position over a given duration. Can optionally move in local space.

```
public static IEnumerator MoveToPosition(this Transform target, Vector3 endPosition, float duration = 0.5, bool useLocal = true, Action onFinished = null)
```

Parameters

target Transform

The Transform to move.

endPosition Vector3

The target position to move towards.

duration float

The duration of the movement in seconds. Defaults to 0.5 seconds.

useLocal bool

If true, operates on localPosition instead of world position.

onFinished Action

An optional callback to invoke when the rotation is complete.

Returns

IEnumerator

An IEnumerator to be used in a coroutine.

Examples

```
StartCoroutine(platformTransform.MoveToPosition(new Vector3(0f, 3f, 0f), 0.8f));
```

MoveToTransform(Transform, Transform, float, bool, Action)

Smoothly moves a Transform to match the position of a target Transform over a specified duration. Can optionally move in local space.

```
public static IEnumerator MoveToTransform(this Transform target, Transform destination,  
float duration = 0.5, bool useLocal = true, Action onFinished = null)
```

Parameters

target Transform

The Transform to move.

destination Transform

The target Transform to move towards.

duration float

The duration of the movement in seconds. Defaults to 0.5 seconds.

useLocal bool

If true, operates on localPosition instead of world position.

onFinished Action

An optional callback to invoke when the rotation is complete.

Returns

IEnumerator

An IEnumerator to be used in a coroutine.

Examples

```
StartCoroutine(enemyTransform.MoveToTransform(playerTransform, 1f, false));
StartCoroutine(handTransform.MoveToTransform(itemSlotTransform, 0.5f));
```

PunchScale(Transform, Vector3, float, int, float)

Applies a punch-scale animation to the transform, simulating an elastic hit effect that decays back to the original scale over the specified duration.

```
public static IEnumerator PunchScale(this Transform transform, Vector3 punch, float duration
= 0.3, int vibrato = 1, float elasticity = 0.5)
```

Parameters

transform Transform

The transform to animate.

punch Vector3

The target scale to punch towards. The animation will start from the current scale and punch towards this value before decaying back.

duration float

Total duration of the animation in seconds.

vibrato int

Number of oscillations over the duration. Higher values produce faster back-and-forth wobbling.

elasticity float

Blends between a simple decay (**0**) and a sinusoidal oscillating decay (**1**).

Returns

IEnumerator

An to be run as a coroutine.

Raycast(Transform, Vector3, out RaycastHit, LayerMask, Vector3, Vector3, float, QueryTriggerInteraction, bool)

Performs a raycast from the Transform's position towards a target position, with optional offsets and settings.

```
public static bool Raycast(this Transform transform, Vector3 targetPosition, out RaycastHit hit, LayerMask includedLayers, Vector3 localRayOffset = null, Vector3 targetRayOffset = null, float maxDistance = 3.4028235E+38, QueryTriggerInteraction queryTriggerInteraction = null, bool drawRay = false)
```

Parameters

transform Transform

The Transform from which the raycast originates.

targetPosition Vector3

The position to raycast towards.

hit RaycastHit

The RaycastHit object that will store information about the hit.

includedLayers LayerMask

The layers to include in the raycast.

localRayOffset Vector3

An optional offset applied to the ray's origin.

targetRayOffset Vector3

An optional offset applied to the ray's target position.

maxDistance float

The maximum distance for the raycast. Defaults to float.MaxValue.

queryTriggerInteraction QueryTriggerInteraction

Specifies whether the raycast should interact with trigger colliders.

drawRay bool

Whether to draw the ray in the scene for debugging purposes.

Returns

bool

True if the raycast hits an object; otherwise, false.

Examples

```
RaycastHit hit;
if(transform.Raycast(target.position, out hit, LayerMask.GetMask("Default")))
{
    Debug.Log("Hit object: " + hit.collider.name);
}
```

RaycastDirection(Transform, Vector3, out RaycastHit, LayerMask, Vector3, float, QueryTriggerInteraction, bool)

Performs a raycast in local space relative to the Transform.


```
public static bool RaycastDirection(this Transform transform, Vector3 localDirection, out RaycastHit hit, LayerMask includedLayers, Vector3 localRayOffset = null, float maxDistance = 3.4028235E+38, QueryTriggerInteraction queryTriggerInteraction = null, bool drawRay = false)
```

Parameters

transform Transform

The Transform from which the raycast originates.

localDirection Vector3

Direction in local space (e.g. Vector3.up, Vector3.forward).

hit RaycastHit

The RaycastHit object that will store information about the hit.

includedLayers LayerMask

The layers to include in the raycast.

localRayOffset Vector3

Optional local space offset applied to the ray's origin.

maxDistance float

The maximum distance for the raycast. Defaults to float.MaxValue.

queryTriggerInteraction QueryTriggerInteraction

Specifies whether the raycast should interact with trigger colliders.

drawRay bool

Whether to draw the ray in the scene for debugging purposes.

Returns

bool

True if the raycast hits an object; otherwise, false.

Examples

```
RaycastHit hit;
if (transform.RaycastDirection(
    Vector3.up,
    out hit,
    LayerMask.GetMask("Default"),
    maxDistance: 10f))
{
    Debug.Log("Hit object above transform: " + hit.collider.name);
}
```

Remarks

Direction and origin offset are transformed into world space internally.

RotateToRotation(Transform, Quaternion, float, bool, Action)

Smoothly rotates a Transform to a specified Quaternion rotation over a given duration. Can optionally rotate in local space.

```
public static IEnumerator RotateToRotation(this Transform target, Quaternion endRot, float
duration = 0.5, bool useLocal = true, Action onFinished = null)
```

Parameters

target Transform

The Transform to rotate.

endRot Quaternion

The target rotation as a Quaternion.

duration float

The duration of the rotation in seconds. Defaults to 0.5 seconds.

useLocal bool

If true, operates on localRotation instead of world rotation.

onFinished Action

An optional callback to invoke when the rotation is complete.

Returns

IEnumerator

An IEnumerator to be used in a coroutine.

Examples

```
StartCoroutine(propellerTransform.RotateToRotation(Quaternion.Euler(0f, 0f, 360f),  
2f, false));  
StartCoroutine(cameraTransform.RotateToRotation(Quaternion.Euler(15f, 0f, 0f), 1f, true));
```

RotateToTransform(Transform, Transform, float, bool, Action)

Smoothly rotates a Transform to match another Transform's rotation over a specified duration. Can optionally rotate in local space.

```
public static IEnumerator RotateToTransform(this Transform target, Transform destination,  
float duration = 0.5, bool useLocal = true, Action onFinished = null)
```

Parameters

target Transform

The Transform to rotate.

destination Transform

The Transform whose rotation to match.

duration float

The duration of the rotation in seconds. Defaults to 0.5 seconds.

useLocal bool

If true, operates on localRotation instead of world rotation.

onFinished Action

An optional callback to invoke when the rotation is complete.

Returns

IEnumerator

An IEnumerator to be used in a coroutine.

Examples

```
StartCoroutine(turretTransform.RotateToTransform(enemyTransform, 0.7f, false));
```

ScaleTo(Transform, Vector3, float, Action)

Smoothly scales a Transform to a target scale over a specified duration.

```
public static IEnumerator ScaleTo(this Transform target, Vector3 endScale, float duration = 0.5, Action onFinished = null)
```

Parameters

target Transform

The Transform to scale.

endScale Vector3

The target local scale.

duration float

Duration of the scaling animation.

onFinished Action

Optional callback when scaling completes.

Returns

IEnumerator

An IEnumerator for coroutine use.

Examples

```
StartCoroutine(transform.ScaleTo(new Vector3(2f, 1f, 1f), 0.3f));
```

Class Vector3Extension

Namespace: MGeLabs.Utils.[Extensions](#)

Assembly: cs.temp.dll.dll

Provides extension methods for the Vector3 class.

```
public static class Vector3Extension
```

Inheritance

object ← Vector3Extension

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

DistanceTo(Vector3, Vector3)

Calculates the distance between two vectors.

```
public static float DistanceTo(this Vector3 vector, Vector3 other)
```

Parameters

vector Vector3

The original vector.

other Vector3

The vector to calculate the distance to.

Returns

float

The distance between the two vectors.

Examples

```
Vector3 a = new Vector3(0, 0, 0);  
Vector3 b = new Vector3(3, 4, 0);  
float dist = a.DistanceTo(b); // Result: 5
```

Divide(Vector3, float)

Divides each component of the vector by a scalar value.

```
public static Vector3 Divide(this Vector3 vector, float value)
```

Parameters

vector Vector3

The original vector.

value float

The scalar value to divide by.

Returns

Vector3

A new vector with each component divided by the scalar value, or the original component if the divisor is zero.

Examples

```
Vector3 v = new Vector3(10, 20, 30);  
Vector3 result = v.Divide(2); // Result: (5, 10, 15)
```

Divide(Vector3, Vector3)

Divides each component of the vector by the corresponding component of another vector.

```
public static Vector3 Divide(this Vector3 vector, Vector3 other)
```

Parameters

vector Vector3

The original vector.

other Vector3

The vector to divide by.

Returns

Vector3

A new vector with each component divided by the corresponding component of the other vector, or the original component if the divisor is zero.

Examples

```
Vector3 a = new Vector3(10, 20, 30);  
Vector3 b = new Vector3(2, 0, 5);  
Vector3 result = a.Divide(b); // Result: (5, 20, 6) - y unchanged because divisor is 0
```

DrawGizmosDownArrow(Vector3, string, Color?, float, float, int, bool)

Draws a downward-pointing gizmo arrow at the given world position (Editor only).

```
public static void DrawGizmosDownArrow(this Vector3 position, string label = "", Color? color = null, float length = 10, float angle = 20, int fontSize = 16, bool drawAdditionalDisk = false)
```

Parameters

position Vector3

World-space position where the arrow tip is placed (arrow points down to this position).

label string

Optional label text shown above the arrow start. Empty string to omit.

color Color?

Optional color for the arrow and label. If null, defaults to .

length float

Distance from the arrow start (top) down to the tip (position).

angle float

Angle (degrees) used to form the arrowhead relative to the shaft.

fontSize int

Font size for the optional label.

drawAdditionalDisk bool

Whether to draw an additional wire disc at the arrow tip for better visibility.

Multiply(Vector3, float)

Multiplies each component of the vector by a scalar value.

```
public static Vector3 Multiply(this Vector3 vector, float value)
```

Parameters

vector Vector3

The original vector.

value float

The scalar value to multiply with.

Returns

Vector3

A new vector with each component multiplied by the scalar value.

Examples

```
Vector3 v = new Vector3(1, 2, 3);  
Vector3 result = v.Multiply(2); // Result: (2, 4, 6)
```

Multiply(Vector3, Vector3)

Multiplies each component of the vector by the corresponding component of another vector.

```
public static Vector3 Multiply(this Vector3 vector, Vector3 other)
```

Parameters

vector Vector3

The original vector.

other Vector3

The vector to multiply with.

Returns

Vector3

A new vector with each component multiplied by the corresponding component of the other vector.

Examples

```
Vector3 a = new Vector3(2, 3, 4);  
Vector3 b = new Vector3(5, 6, 7);  
Vector3 result = a.Multiply(b); // Result: (10, 18, 28)
```

With(Vector3, float?, float?, float?)

Creates a new vector with the specified components replaced by the provided values.

```
public static Vector3 With(this Vector3 vector, float? x = null, float? y = null, float? z = null)
```

Parameters

vector Vector3

The original vector.

x float?

The new x-component value, or null to keep the original x-component.

y float?

The new y-component value, or null to keep the original y-component.

z float?

The new z-component value, or null to keep the original z-component.

Returns

Vector3

A new vector with the specified components replaced by the provided values.

Examples

```
Vector3 v = new Vector3(1, 2, 3);  
Vector3 modified = v.With(y: 10); // Result: (1, 10, 3)
```

Namespace MGeLabs.Utils.FSM

Classes

[FunctionPredicate](#)

A concrete implementation of [IPredicate](#) that uses a function delegate to evaluate a condition.

[NotPredicate](#)

Represents a predicate that negates the result of another predicate.

[StateMachine](#)

Represents a state machine that manages state transitions and updates.

[StateNode](#)

Represents a node in the state machine, holding a state and its transitions.

[Transition](#)

Implements a transition between states in a state machine.

Interfaces

[IPredicate](#)

Represents a predicate that can be evaluated to determine a boolean condition.

[IState](#)

Represents a state in the state machine.

[ITransition](#)

Represents a transition between states in a state machine.

Class FunctionPredicate

Namespace: MGeLabs.Utils.[FSM](#)

Assembly: cs.temp.dll.dll

A concrete implementation of [IPredicate](#) that uses a function delegate to evaluate a condition.

```
public class FunctionPredicate : IPredicate
```

Inheritance

object ← FunctionPredicate

Implements

[IPredicate](#)

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Examples

Example usage:

```
bool isHealthLow = false;
IPredicate lowHealthPredicate = new FunctionPredicate(() => isHealthLow);

if (lowHealthPredicate.Evaluate())
{
    Debug.Log("Health is low!");
}
```

Constructors

FunctionPredicate(Func<bool>)

```
public FunctionPredicate(Func<bool> predicate)
```

Parameters

`predicate` Func<bool>

Methods

Evaluate()

```
public bool Evaluate()
```

Returns

bool

Operators

operator !(FunctionPredicate)

```
public static IPredicate operator !(FunctionPredicate original)
```

Parameters

`original` [FunctionPredicate](#)

Returns

[IPredicate](#)

Interface IPredicate

Namespace: MGeLabs.Utils.[FSM](#)

Assembly: cs.temp.dll.dll

Represents a predicate that can be evaluated to determine a boolean condition.

```
public interface IPredicate
```

Methods

Evaluate()

```
bool Evaluate()
```

Returns

bool

Interface IState

Namespace: MGeLabs.Utils.[FSM](#)

Assembly: cs.temp.dll.dll

Represents a state in the state machine.

```
public interface IState
```

Examples

Example implementation of a state:

```
public class IdleState : IState
{
    public void Enter()
    {
        Debug.Log("Entering Idle State");
    }

    public void Update()
    {
        // Idle behavior
    }

    public void FixedUpdate()
    {
        // Physics-related idle behavior
    }

    public void Exit()
    {
        Debug.Log("Exiting Idle State");
    }
}
```

Methods

Enter()

Called when the state is entered.

```
void Enter()
```

Exit()

Called when the state is exited.

```
void Exit()
```

FixedUpdate()

Called at fixed intervals to update the state.

```
void FixedUpdate()
```

Update()

Called every frame to update the state.

```
void Update()
```

Interface ITransition

Namespace: MGeLabs.Utils.[FSM](#)

Assembly: cs.temp.dll.dll

Represents a transition between states in a state machine.

```
public interface ITransition
```

Properties

Predicate

Gets the predicate that determines if the transition should occur.

```
IPredicate Predicate { get; }
```

Property Value

[IPredicate](#)

TargetState

Gets the target state of the transition.

```
IState TargetState { get; }
```

Property Value

[IState](#)

Class NotPredicate

Namespace: MGeLabs.Utils.[FSM](#)

Assembly: cs.temp.dll.dll

Represents a predicate that negates the result of another predicate.

```
public class NotPredicate : IPredicate
```

Inheritance

object ← NotPredicate

Implements

[IPredicate](#)

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

NotPredicate(IPredicate)

```
public NotPredicate(IPredicate original)
```

Parameters

original [IPredicate](#)

Methods

Evaluate()

Evaluates the predicate by negating the result of the original predicate.

```
public bool Evaluate()
```

Returns

bool

`true` if the original predicate evaluates to `false`; otherwise, `false`.

Class StateMachine

Namespace: MGeLabs.Utils.[FSM](#)

Assembly: cs.temp.dll.dll

Represents a state machine that manages state transitions and updates.

```
public class StateMachine
```

Inheritance

object ← StateMachine

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Examples

Example usage:

```
public class IdleState : IState
{
    public void Enter() => Debug.Log("Entered Idle");
    public void Update() => Debug.Log("Updating Idle");
    public void FixedUpdate() { }
    public void Exit() => Debug.Log("Exited Idle");
}

public class MoveState : IState
{
    public void Enter() => Debug.Log("Entered Move");
    public void Update() => Debug.Log("Updating Move");
    public void FixedUpdate() { }
    public void Exit() => Debug.Log("Exited Move");
}

void SetupStateMachine()
{
    IdleState idle = new IdleState();
    MoveState move = new MoveState();
    StateMachine stateMachine = new StateMachine();
}
```

```

// Add transitions
stateMachine.AddTransition(idle, move, new FunctionPredicate(()
=> Input.GetKey(KeyCode.Space)));
stateMachine.AddTransition(move, idle, new FunctionPredicate(() =>
!Input.GetKey(KeyCode.Space)));

// Start in Idle
stateMachine.SetState(idle);

// In your MonoBehaviour.Update():
stateMachine.Update();
}

```

Properties

CurrentState

```
public StateNode CurrentState { get; }
```

Property Value

[StateNode](#)

Methods

AddAnyTransition(IState, IPredicate)

Adds a transition from any state to a specified state with a specified predicate.

```
public void AddAnyTransition(IState to, IPredicate predicate)
```

Parameters

to [IState](#)

The state to transition to.

predicate [IPredicate](#)

The predicate that determines if the transition should occur.

AddTransition(IState, IState, IPredicate)

Adds a transition from one state to another with a specified predicate.

```
public void AddTransition(IState from, IState to, IPredicate predicate)
```

Parameters

from [IState](#)

The state to transition from.

to [IState](#)

The state to transition to.

predicate [IPredicate](#)

The predicate that determines if the transition should occur.

FixedUpdate()

Calls the FixedUpdate method on the current state.

```
public void FixedUpdate()
```

SetState(IState)

Sets the current state of the state machine.

```
public void SetState(IState state)
```

Parameters

state [IState](#)

The new state to set.

Update()

Updates the state machine, checking for transitions and updating the current state.

```
public void Update()
```

Events

OnStateChanged

```
public event Action<IState> OnStateChanged
```

Event Type

Action<[IState](#)>

Class StateNode

Namespace: MGeLabs.Utils.[FSM](#)

Assembly: cs.temp.dll.dll

Represents a node in the state machine, holding a state and its transitions.

```
public sealed class StateNode
```

Inheritance

object $\leftarrow$ StateNode

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

StateNode(IState)

Initializes a new instance of the [StateNode](#) class with the specified state.

```
public StateNode(IState state)
```

Parameters

state [IState](#)

The state to associate with this node.

Properties

State

Gets the state associated with this node.

```
public IState State { get; }
```

Property Value

[IState](#)

Transitions

Gets the set of transitions from this state.

```
public HashSet<ITransition> Transitions { get; }
```

Property Value

HashSet<[ITransition](#)>

Methods

AddTransition(IState, IPredicate)

Adds a transition from this state to another state with a specified predicate.

```
public void AddTransition(IState to, IPredicate predicate)
```

Parameters

to [IState](#)

The state to transition to.

predicate [IPredicate](#)

The predicate that determines if the transition should occur.

Class Transition

Namespace: MGeLabs.Utils.[FSM](#)

Assembly: cs.temp.dll.dll

Implements a transition between states in a state machine.

```
public class Transition : ITransition
```

Inheritance

object ← Transition

Implements

[ITransition](#)

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Examples

Example usage:

```
void CreateTransition()
{
    IState idle = new IdleState();
    IState move = new MoveState();

    IPredicate canMove = new FunctionPredicate(() => Input.GetKey(KeyCode.Space));

    ITransition transition = new Transition(canMove, move);

    // Example: use in a state machine
    StateMachine stateMachine = new StateMachine();
    stateMachine.AddTransition(idle, move, canMove);
}
```

Constructors

Transition(IPredicate, IState)

Initializes a new instance of the [Transition](#) class.

```
public Transition(IPredicate predicate, IState targetState)
```

Parameters

predicate [IPredicate](#)

The predicate that determines if the transition should occur.

targetState [IState](#)

The target state of the transition.

Properties

Predicate

Gets the predicate that determines if the transition should occur.

```
public IPredicate Predicate { get; }
```

Property Value

[IPredicate](#)

TargetState

Gets the target state of the transition.

```
public IState TargetState { get; }
```

Property Value

[IState](#)

Namespace MGeLabs.Utils.GameObjects

Classes

[CacheDictionary<TKey, T>](#)

Represents a dictionary that caches items and provides methods to manage the cache.

[ColliderEventHandler](#)

Handles runtime registration of collision and trigger events based on object tags.

[DestroyAfterTime](#)

A utility class that destroys or disables the GameObject after a specified lifetime. Triggers UnityEvent and Action callbacks before destruction.

[EphemeralSingleton<T>](#)

An ephemeral singleton that does not persist across scene loads.

[GenericObjectPool<T>](#)

A generic object pool for managing reusable objects of type T.

[PersistentObject](#)

A Unity component that assigns a unique GUID to the object for persistent identification.

[PersistentSingleton<T>](#)

A persistent singleton that is not destroyed on scene load.

[Singleton<T>](#)

Abstract base class for creating singleton MonoBehaviour instances.

[TransformAnimator](#)

A component that animates the position, rotation, and scale of a target Transform based on configurable animation settings.

[TransformAnimator.TransformAnimatorComponentData](#)

Represents the configuration data for animating a specific transform component.

[TransformFollower](#)

A simple helper component that follows a target in world space. Attach to any GameObject to make it move (and optionally rotate) towards another transform.

Structs

[ColliderEvent](#)

Struct representing a collider event with callbacks for enter and exit interactions.

[ColliderEventEntry](#)

Stores collision and trigger events.

Interfaces

[IPoolable](#)

Interface for objects that can be managed by a pool. Provides methods to handle pooling, recycling, and returning operations.

Class CacheDictionary<TKey, T>

Namespace: MGeLabs.Utils.[GameObjects](#)

Assembly: cs.temp.dll.dll

Represents a dictionary that caches items and provides methods to manage the cache.

```
public abstract class CacheDictionary<TKey, T> : EphemeralSingleton<CacheDictionary<TKey, T>>
```

Type Parameters

TKey

The type of the key used in the dictionary.

T

The type of the value stored in the dictionary.

Inheritance

object ← [Singleton](#) <[CacheDictionary](#) <TKey, T>> ← [EphemeralSingleton](#) <[CacheDictionary](#) <TKey, T>> ← [CacheDictionary](#) <TKey, T>

Inherited Members

[Singleton](#) <[CacheDictionary](#) <TKey, T>>.instance , [Singleton](#) <[CacheDictionary](#) <TKey, T>>.Instance , [Singleton](#) <[CacheDictionary](#) <TKey, T>>.Awake()

Examples

Example of using a cache to store surface types based on a ground transform:

```
public class SurfaceType
{
    public string type;
}

public class GroundSurfaceCache : CacheDictionary<Transform, string> { }

public class Player
{
    public void OnFootstep(Transform groundTransform)
```

```

{
    string surfaceType = GroundSurfaceCache.Instance.GetOrAdd(
        groundTransform,
        transform => transform.GetComponent<SurfaceType>()?.type,
        discardNullValue: true
    );
    Debug.Log("Footstep on surface: " + surfaceType);
}
}

```

Methods

AddCachedItem(TKey, T)

Adds or updates an item in the cache.

```
public void AddCachedItem(TKey key, T value)
```

Parameters

key TKey

The key of the item to add or update.

value T

The value to associate with the key.

ClearCache()

Clears all items from the cache.

```
public void ClearCache()
```

ContainsKey(TKey)

Determines whether the cache contains the specified key.


```
public bool ContainsKey(TKey key)
```

Parameters

key TKey

The key to check for existence.

Returns

bool

true if the key exists in the cache; otherwise, **false**.

GetOrAdd(TKey, Func<TKey, T>, bool)

Retrieves a value from the cache or adds it using the specified factory function if it does not exist.

```
public T GetOrAdd(TKey key, Func<TKey, T> valueFactory, bool discardNullValue = true)
```

Parameters

key TKey

The key to retrieve or add.

valueFactory Func<TKey, T>

The function used to create the value if it does not exist.

discardNullValue bool

Indicates whether to discard null values. Defaults to **true**.

Returns

T

The value associated with the key.

RemoveCachedItem(TKey)

Removes an item from the cache.

```
public void RemoveCachedItem(TKey key)
```

Parameters

key TKey

The key of the item to remove.

TryGetValue(TKey, out T)

Attempts to retrieve a value from the cache.

```
public bool TryGetValue(TKey key, out T value)
```

Parameters

key TKey

The key to retrieve.

value T

When this method returns, contains the value associated with the key if found; otherwise, the default value for the type of the value parameter.

Returns

bool

true if the key exists in the cache; otherwise, **false**.

Struct ColliderEvent

Namespace: MGeLabs.Utils.[GameObjects](#)

Assembly: cs.temp.dll.dll

Struct representing a collider event with callbacks for enter and exit interactions.

```
public struct ColliderEvent
```

Inherited Members

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Fields

OnEnter

The action to invoke when a collider with the matching tag enters this collider.

```
public Action<GameObject, ContactPoint[]> OnEnter
```

Field Value

Action<GameObject, ContactPoint[]>

Remarks

The first parameter is the other UnityEngine.GameObject involved in the collision. The second parameter is an array of UnityEngine.ContactPoint providing collision details.

OnExit

The action to invoke when a collider with the matching tag exits this collider.

```
public Action<GameObject, ContactPoint[]> OnExit
```

Field Value

Action<GameObject, ContactPoint[]>

Remarks

The first parameter is the other UnityEngine.GameObject involved in the collision. The second parameter is an array of UnityEngine.ContactPoint from the exit interaction.

Tag

The tag of the GameObject to match during collision events. Only GameObjects with this tag will trigger the callbacks.

```
public string Tag
```

Field Value

string

Struct ColliderEventEntry

Namespace: MGeLabs.Utils.[GameObjects](#)

Assembly: cs.temp.dll.dll

Stores collision and trigger events.

```
public struct ColliderEventEntry
```

Inherited Members

ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Fields

CollisionEvent

```
public ColliderEvent CollisionEvent
```

Field Value

[ColliderEvent](#)

TriggerEvent

```
public ColliderEvent TriggerEvent
```

Field Value

[ColliderEvent](#)

Class ColliderEventHandler

Namespace: MGeLabs.Utils.[GameObjects](#)

Assembly: cs.temp.dll.dll

Handles runtime registration of collision and trigger events based on object tags.

```
public abstract class ColliderEventHandler : MonoBehaviour
```

Inheritance

object ← ColliderEventHandler

Examples

```
public class MyColliderHandler : ColliderEventHandler
{
    private void Start()
    {
        // Register a collision event for objects tagged "Enemy"
        RegisterColliderEvent(new ColliderEvent
        {
            Tag = "Enemy",
            OnEnter = (go, contacts) => Debug.Log($"Enemy collided with {go.name}"),
            OnExit = (go, contacts) => Debug.Log($"Enemy stopped colliding with {go.name}")
        });

        // Register a trigger event for objects tagged "Pickup"
        RegisterColliderEvent(new ColliderEvent
        {
            Tag = "Pickup",
            OnEnter = (go, _) => Debug.Log($"Entered pickup trigger: {go.name}"),
            OnExit = (go, _) => Debug.Log($"Exited pickup trigger: {go.name}")
        }, isTriggerEvent: true);
    }
}
```

Fields

registeredEvents

Stores collision and trigger events by tag.

```
protected readonly Dictionary<string, ColliderEventEntry> registeredEvents
```

Field Value

Dictionary<string, [ColliderEventEntry](#)>

Methods

OnCollisionEnter(Collision)

Called when a collision occurs.

```
protected virtual void OnCollisionEnter(Collision other)
```

Parameters

other Collision

The collision data associated with the collision event.

OnCollisionExit(Collision)

Called when a collision ends.

```
protected virtual void OnCollisionExit(Collision other)
```

Parameters

other Collision

The collision data associated with the collision event.

OnTriggerEnter(Collider)

Called when a trigger collider enters the trigger.

```
protected virtual void OnTriggerEnter(Collider other)
```

Parameters

other Collider

The other collider involved in the trigger event.

OnTriggerExit(Collider)

Called when a trigger collider exits the trigger.

```
protected virtual void OnTriggerExit(Collider other)
```

Parameters

other Collider

The other collider involved in the trigger event.

RegisterColliderEvent(ColliderEvent, bool)

Registers a collider event by tag.

```
protected virtual void RegisterColliderEvent(ColliderEvent colliderEvent, bool  
isTriggerEvent = false)
```

Parameters

colliderEvent [ColliderEvent](#)

isTriggerEvent bool

UnregisterColliderEvent(string)

Unregisters a collider event by a given tag.


```
protected virtual void UnregisterColliderEvent(string eventTag)
```

Parameters

eventTag string

Class DestroyAfterTime

Namespace: MGeLabs.Utils.[GameObjects](#)

Assembly: cs.temp.dll.dll

A utility class that destroys or disables the GameObject after a specified lifetime. Triggers UnityEvent and Action callbacks before destruction.

```
public class DestroyAfterTime : MonoBehaviour
```

Inheritance

object ← DestroyAfterTime

Fields

disableOnly

```
protected bool disableOnly
```

Field Value

bool

lifetime

```
protected float lifetime
```

Field Value

float

OnDestroy

UnityEvent invoked before the GameObject is destroyed.

```
public UnityEvent OnDestroy
```

Field Value

UnityEvent

Methods

DestroySelf()

Invokes the destruction callbacks and destroys or disables the GameObject.

```
public virtual void DestroySelf()
```

OnDisable()

Called when the GameObject is disabled. Cancels the scheduled destruction of the GameObject.

```
protected virtual void OnDisable()
```

OnEnable()

Called when the GameObject is enabled. Schedules the destruction of the GameObject.

```
protected virtual void OnEnable()
```

SetLifetime(float)

Sets the lifetime used for scheduled destruction and, if this component is enabled, reschedules the pending destruction with the new lifetime.

```
public virtual void SetLifetime(float newLifetime)
```

Parameters

`newLifetime` float

New lifetime in seconds.

Events

OnDestroyAction

Action event invoked before the GameObject is destroyed.

```
public event Action OnDestroyAction
```

Event Type

Action

Class EphemeralSingleton<T>

Namespace: MGeLabs.Utils.[GameObjects](#)

Assembly: cs.temp.dll.dll

An ephemeral singleton that does not persist across scene loads.

```
public abstract class EphemeralSingleton<T> : Singleton<T> where T : MonoBehaviour
```

Type Parameters

T

Type of the singleton class.

Inheritance

object ← [Singleton](#)<T> ← EphemeralSingleton<T>

Derived

[CacheDictionary<TKey, T>](#), [GenericObjectPool<T>](#)

Inherited Members

[Singleton<T>.instance](#), [Singleton<T>.Instance](#), [Singleton<T>.Awake\(\)](#)

Examples

```
// Example usage of EphemeralSingleton
public class UIManager : EphemeralSingleton<UIManager>
{
    public void ShowMenu()
    {
        Debug.Log("Showing menu");
    }
}

public class MainMenu : MonoBehaviour
{
    void OnEnable()
    {
        UIManager.Instance.ShowMenu();
    }
}
```

}
}

Class GenericObjectPool<T>

Namespace: MGeLabs.Utils.[GameObjects](#)

Assembly: cs.temp.dll.dll

A generic object pool for managing reusable objects of type T.

```
public abstract class GenericObjectPool<T> : EphemeralSingleton<GenericObjectPool<T>> where  
T : Component
```

Type Parameters

T

The type of objects to pool, which must be a Component.

Inheritance

object ← [Singleton](#)<[GenericObjectPool](#)<T>> ← [EphemeralSingleton](#)<[GenericObjectPool](#)<T>> ←
[GenericObjectPool](#)<T>

Inherited Members

[Singleton](#)<[GenericObjectPool](#)<T>>.instance , [Singleton](#)<[GenericObjectPool](#)<T>>.Instance

Fields

activatePooledObjects

```
protected bool activatePooledObjects
```

Field Value

bool

activeObjects

```
protected readonly LinkedList<T> activeObjects
```

Field Value

LinkedList<T>

allObjects

```
protected readonly HashSet<T> allObjects
```

Field Value

HashSet<T>

allowObjectRecycling

```
protected bool allowObjectRecycling
```

Field Value

bool

awakeObjectsBatchSize

```
protected int awakeObjectsBatchSize
```

Field Value

int

awakeObjectsOnCreation

```
protected bool awakeObjectsOnCreation
```

Field Value

bool

maxPoolSize

```
protected int maxPoolSize
```

Field Value

int

prefab

```
protected T prefab
```

Field Value

T

prewarmCount

```
protected int prewarmCount
```

Field Value

int

readyToUseObjects

```
protected readonly Queue<T> readyToUseObjects
```

Field Value

Queue<T>

Properties

ActiveObjects

```
public IReadOnlyCollection<T> ActiveObjects { get; }
```

Property Value

`IReadOnlyCollection<T>`

The collection of currently active objects in the pool.

AvailableSlots

```
public int AvailableSlots { get; }
```

Property Value

`int`

The number of available slots in the pool for new objects.

Methods

AddObjects(int)

Adds a specified number of objects to the pool, up to the available slots.

```
protected virtual void AddObjects(int count)
```

Parameters

`count` `int`

The number of objects to add to the pool.

Awake()

Initializes the pool and prewarms objects if specified.

```
protected override void Awake()
```

Examples

```
public class Bullet : MonoBehaviour
{
    // Bullet logic here
}

public class BulletPool : GenericObjectPool<Bullet>
{
    // Optionally add pool-specific logic here
    // The Pool has to be placed in the scene. Within the inspector, you have to assign
    // it's prefab.
}

public class PlayerShooter : MonoBehaviour
{
    private void Shoot()
    {
        Bullet bullet = BulletPool.Instance.Get();
        bullet.transform.position = transform.position;
        bullet.gameObject.SetActive(true);

        // Setup bullet, e.g., velocity, direction...
    }

    private void OnBulletFinished(Bullet bullet)
    {
        BulletPool.Instance.ReturnToPool(bullet);
    }
}
```

CreateInstance()

Creates a new instance of the prefab and initializes it for use in the pool.

```
protected virtual T CreateInstance()
```

Returns

T

A new instance of type T with its GameObject deactivated.

Get()

Retrieves an object from the pool. If no objects are available, attempts to create or recycle one.

```
public virtual T Get()
```

Returns

T

A pooled object of type T or null if the operation fails.

Examples

```
// Example: shooting bullets
Bullet bullet = BulletPool.Instance.Get();
if (bullet != null)
{
    bullet.transform.position = transform.position;
}
```

Get(Vector3, Quaternion, Transform)

Retrieves an object from the pool and sets its position, rotation, and parent transform. If no objects are available, attempts to create or recycle one.

```
public virtual T Get(Vector3 position, Quaternion rotation, Transform parent = null)
```

Parameters

position Vector3

The position to set for the object.

rotation Quaternion

The rotation to set for the object.

parent Transform

The parent transform to assign to the object. Defaults to **null**.

Returns

T

A pooled object of type **T** or **null** if the operation fails.

Examples

```
Vector3 pos = transform.position + transform.forward * 2f;  
Quaternion rot = Quaternion.identity;  
Bullet bullet = BulletPool.Instance.Get(pos, rot, this.transform);
```

GetOrCreateObject(out bool)

Retrieves an object from the pool or creates/recycles one if necessary.

```
protected virtual T GetOrCreateObject(out bool wasNew)
```

Parameters

wasNew bool

Outputs whether the object was newly created (**true**) or retrieved from the pool (**false**).

Returns

T

A pooled object of type **T** or **null** if the operation fails.

NotifyPooled(T, bool)

Notifies that an object has been pooled and invokes relevant events.

```
protected virtual void NotifyPooled(T obj, bool wasNew)
```

Parameters

obj T

The object that has been pooled.

wasNew bool

Indicates whether the object was newly created (**true**) or retrieved from the pool (**false**).

PrewarmCoroutine(int)

Prewarms the object pool by activating and deactivating objects in batches. This ensures that objects are properly initialized before use.

```
protected virtual IEnumerator PrewarmCoroutine(int batchSize = 50)
```

Parameters

batchSize int

The number of objects to activate and deactivate in each batch. Defaults to 50.

Returns

IEnumerator

An enumerator that performs the prewarming operation over multiple frames.

ReturnToPool(T)

Returns an object to the pool.

```
public virtual void ReturnToPool(T objectToReturn)
```

Parameters

objectToReturn T

The object to return to the pool.

Examples

```
// When a bullet finishes:  
BulletPool.Instance.ReturnToPool(bullet);
```

Events

OnObjectPooled

```
public event Action<T> OnObjectPooled
```

Event Type

Action<T>

OnObjectRecycled

```
public event Action<T> OnObjectRecycled
```

Event Type

Action<T>

OnObjectReturned

```
public event Action<T> OnObjectReturned
```

Event Type

Action<T>

OnPrewarmCompleted

```
public event Action<List<T>> OnPrewarmCompleted
```

Event Type

Action<List<T>>

Interface IPoolable

Namespace: MGeLabs.Utils.[GameObjects](#)

Assembly: cs.temp.dll.dll

Interface for objects that can be managed by a pool. Provides methods to handle pooling, recycling, and returning operations.

```
public interface IPoolable
```

Methods

OnPooled()

```
void OnPooled()
```

OnRecycled()

```
void OnRecycled()
```

OnReturned()

```
void OnReturned()
```

Class PersistentObject

Namespace: MGeLabs.Utils.[GameObjects](#)

Assembly: cs.temp.dll.dll

A Unity component that assigns a unique GUID to the object for persistent identification.

```
public class PersistentObject : MonoBehaviour
```

Inheritance

object ← PersistentObject

Examples

```
// Attach PersistentObject component to any GameObject in the editor.  
// When you create or duplicate the GameObject, it automatically gets a unique GUID.
```

```
public class ExampleUsage : MonoBehaviour  
{  
    private void Start()  
    {  
        PersistentObject persistent = gameObject.GetComponent<PersistentObject>();  
        if (persistent != null)  
        {  
            Debug.Log("Persistent GUID: " + persistent.guid);  
        }  
    }  
}
```

Fields

guid

```
public string guid
```

Field Value

string

Class PersistentSingleton<T>

Namespace: MGeLabs.Utils.[GameObjects](#)

Assembly: cs.temp.dll.dll

A persistent singleton that is not destroyed on scene load.

```
public abstract class PersistentSingleton<T> : Singleton<T> where T : MonoBehaviour
```

Type Parameters

T

Type of the singleton class.

Inheritance

object ← [Singleton](#)<T> ← PersistentSingleton<T>

Inherited Members

[Singleton<T>.instance](#) , [Singleton<T>.Instance](#)

Examples

```
// Example usage of PersistentSingleton
public class AudioManager : PersistentSingleton<AudioManager>
{
    public void PlaySound(string clipName)
    {
        Debug.Log("Playing sound: " + clipName);
    }
}

public class GameController : MonoBehaviour
{
    void Start()
    {
        AudioManager.Instance.PlaySound("GameStart");
    }
}
```

Methods

Awake()

Awake method to call DontDestroyOnLoad on the object.

```
protected override void Awake()
```

Class Singleton<T>

Namespace: MGeLabs.Utils.[GameObjects](#)

Assembly: cs.temp.dll.dll

Abstract base class for creating singleton MonoBehaviour instances.

```
public abstract class Singleton<T> : MonoBehaviour where T : MonoBehaviour
```

Type Parameters

T

Type of the singleton class.

Inheritance

object ← Singleton<T>

Derived

[EphemeralSingleton<T>](#), [PersistentSingleton<T>](#)

Fields

instance

```
protected static T instance
```

Field Value

T

Properties

Instance

Gets the singleton instance. If the instance is not found, it searches for it in the scene.

```
public static T Instance { get; }
```

Property Value

T

The singleton instance of type T.

Methods

Awake()

Awake method to initialize the singleton instance.

```
protected virtual void Awake()
```

Class TransformAnimator

Namespace: MGeLabs.Utils.[GameObjects](#)

Assembly: cs.temp.dll.dll

A component that animates the position, rotation, and scale of a target Transform based on configurable animation settings.

```
public class TransformAnimator : MonoBehaviour
```

Inheritance

object ← TransformAnimator

Fields

hasCompletedPositionCycle

```
protected bool hasCompletedPositionCycle
```

Field Value

bool

hasCompletedRotationCycle

```
protected bool hasCompletedRotationCycle
```

Field Value

bool

hasCompletedScaleCycle

`protected bool` hasCompletedScaleCycle

Field Value

bool

isEnabled

`protected bool` isEnabled

Field Value

bool

isPositionForward

`protected bool` isPositionForward

Field Value

bool

isRotationForward

`protected bool` isRotationForward

Field Value

bool

isScaleForward

`protected bool isScaleForward`

Field Value

bool

OnPositionTransitionCompleted

`public UnityEvent OnPositionTransitionCompleted`

Field Value

UnityEvent

OnRotationTransitionCompleted

`public UnityEvent OnRotationTransitionCompleted`

Field Value

UnityEvent

OnScaleTransitionCompleted

`public UnityEvent OnScaleTransitionCompleted`

Field Value

UnityEvent

positionConfig

`protected TransformAnimator.TransformAnimatorComponentData positionConfig`

Field Value

[TransformAnimator.TransformAnimatorComponentData](#)

positionDelayTimer

`protected float positionDelayTimer`

Field Value

float

positionProgress

`protected float positionProgress`

Field Value

float

positionTarget

`protected Vector3 positionTarget`

Field Value

Vector3

rotationConfig

`protected TransformAnimator.TransformAnimatorComponentData rotationConfig`

Field Value

[TransformAnimator.TransformAnimatorComponentData](#)

rotationDelayTimer

`protected float rotationDelayTimer`

Field Value

float

rotationProgress

`protected float rotationProgress`

Field Value

float

rotationTarget

`protected Vector3 rotationTarget`

Field Value

Vector3

scaleConfig

`protected TransformAnimator.TransformAnimatorComponentData scaleConfig`

Field Value

[TransformAnimator.TransformAnimatorComponentData](#)

scaleDelayTimer

`protected float scaleDelayTimer`

Field Value

float

scaleProgress

`protected float scaleProgress`

Field Value

float

scaleTarget

`protected Vector3 scaleTarget`

Field Value

Vector3

target

`protected` Transform target

Field Value

Transform

Properties

PositionConfig

```
public TransformAnimator.TransformAnimatorComponentData PositionConfig { get; }
```

Property Value

[TransformAnimator.TransformAnimatorComponentData](#)

RotationConfig

```
public TransformAnimator.TransformAnimatorComponentData RotationConfig { get; }
```

Property Value

[TransformAnimator.TransformAnimatorComponentData](#)

ScaleConfig

```
public TransformAnimator.TransformAnimatorComponentData ScaleConfig { get; }
```

Property Value

[TransformAnimator.TransformAnimatorComponentData](#)

Methods

AnimatePosition()

Animates the position of the target Transform based on the configuration.

```
protected virtual void AnimatePosition()
```

AnimateRotation()

Animates the rotation of the target Transform based on the configuration.

```
protected virtual void AnimateRotation()
```

AnimateScale()

Animates the scale of the target Transform based on the configuration.

```
protected virtual void AnimateScale()
```

Awake()

Initializes the target Transform and sets its initial position, rotation, and scale based on the animation configurations. Also calculates randomized targets for animations.

```
protected virtual void Awake()
```

GetRandomizedTarget(Vector3, Vector3)

Calculates a randomized target value by applying a random offset to the base end value.

```
protected static Vector3 GetRandomizedTarget(Vector3 baseEnd, Vector3 offset)
```

Parameters

baseEnd Vector3

The base end value to which the random offset will be applied.

`offset` Vector3

The range of random offset to apply to each axis.

Returns

Vector3

A new Vector3 with randomized values within the specified offset range.

PlayBackward()

Sets the animation direction to backward for position, rotation, and scale.

```
public virtual void PlayBackward()
```

PlayForward()

Sets the animation direction to forward for position, rotation, and scale.

```
public virtual void PlayForward()
```

ResetProgress()

Resets the progress and delay timers for position, rotation, and scale animations.

```
public virtual void ResetProgress()
```

SetIsEnabled(bool)

Enables or disables the animator.

```
public virtual void SetIsEnabled(bool state)
```

Parameters

state bool

True to enable, false to disable.

SetPositionConfig(TransformAnimatorComponentData)

Sets the configuration for position animation.

```
public virtual void SetPositionConfig(TransformAnimator.TransformAnimatorComponentData config)
```

Parameters

config [TransformAnimator.TransformAnimatorComponentData](#)

The configuration data for position animation.

SetRotationConfig(TransformAnimatorComponentData)

Sets the configuration for rotation animation.

```
public virtual void SetRotationConfig(TransformAnimator.TransformAnimatorComponentData config)
```

Parameters

config [TransformAnimator.TransformAnimatorComponentData](#)

The configuration data for rotation animation.

SetScaleConfig(TransformAnimatorComponentData)

Sets the configuration for scale animation.

```
public virtual void SetScaleConfig(TransformAnimator.TransformAnimatorComponentData config)
```


Parameters

config [TransformAnimator.TransformAnimatorComponentData](#)

The configuration data for scale animation.

Update()

Updates the animation for position, rotation, and scale on each frame.

```
protected virtual void Update()
```

Events

OnPositionTransitionCompletedAction

```
public event Action OnPositionTransitionCompletedAction
```

Event Type

Action

OnRotationTransitionCompletedAction

```
public event Action OnRotationTransitionCompletedAction
```

Event Type

Action

OnScaleTransitionCompletedAction

```
public event Action OnScaleTransitionCompletedAction
```

Event Type

Action

Class TransformAnimator.TransformAnimatorComponentData

Namespace: MGeLabs.Utils.[GameObjects](#)

Assembly: cs.temp.dll.dll

Represents the configuration data for animating a specific transform component.

```
public class TransformAnimator.TransformAnimatorComponentData
```

Inheritance

object ← TransformAnimator.TransformAnimatorComponentData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

animationCurve

```
public AnimationCurve animationCurve
```

Field Value

AnimationCurve

canLoop

```
public bool canLoop
```

Field Value

bool

endValue

```
public Vector3 endValue
```

Field Value

Vector3

isContinuous

```
public bool isContinuous
```

Field Value

bool

isEnabled

```
public bool isEnabled
```

Field Value

bool

loopDelay

```
public float loopDelay
```

Field Value

float

speed

```
public float speed
```

Field Value

float

startValue

```
public Vector3 startValue
```

Field Value

Vector3

targetVariance

```
public Vector3 targetVariance
```

Field Value

Vector3

Class TransformFollower

Namespace: MGeLabs.Utils.[GameObjects](#)

Assembly: cs.temp.dll.dll

A simple helper component that follows a target in world space. Attach to any GameObject to make it move (and optionally rotate) towards another transform.

```
public class TransformFollower : MonoBehaviour
```

Inheritance

object ← TransformFollower

Fields

followPosition

```
protected bool followPosition
```

Field Value

bool

followRotation

```
protected bool followRotation
```

Field Value

bool

followSpeed

`protected float followSpeed`

Field Value

float

positionOffset

`protected Vector3 positionOffset`

Field Value

Vector3

rotationSpeed

`protected float rotationSpeed`

Field Value

float

target

`protected Transform target`

Field Value

Transform

Methods

FollowPosition()

Moves this follower towards the target's position while applying the configured offset and speed.

```
protected virtual void FollowPosition()
```

FollowRotation()

Rotates this follower towards the target's rotation at the configured angular speed.

```
protected virtual void FollowRotation()
```

SetCanFollowPosition(bool)

Enable or disable position following.

```
public TransformFollower SetCanFollowPosition(bool canFollow)
```

Parameters

canFollow bool

If true, the follower will move to match the target's position; if false, position following is disabled.

Returns

[TransformFollower](#)

The current [TransformFollower](#) instance for fluent/chaining calls.

SetCanFollowRotation(bool)

Enable or disable rotation following.

```
public TransformFollower SetCanFollowRotation(bool canFollow)
```

Parameters

`canFollow` bool

If true, the follower will rotate to match the target.

Returns

[TransformFollower](#)

This instance for fluent/chaining calls.

SetFollowSpeed(float)

Set the follow speed (units per second).

```
public TransformFollower SetFollowSpeed(float newSpeed)
```

Parameters

`newSpeed` float

Linear follow speed in units/second.

Returns

[TransformFollower](#)

This instance for fluent/chaining calls.

SetPositionOffset(Vector3)

Set the world-space position offset relative to the target.

```
public TransformFollower SetPositionOffset(Vector3 newOffset)
```

Parameters

`newOffset` Vector3

Offset added to the target's position.

Returns

[TransformFollower](#)

This instance for fluent/chaining calls.

SetRotationSpeed(float)

Set the rotation speed (degrees per second).

```
public TransformFollower SetRotationSpeed(float newSpeed)
```

Parameters

newSpeed float

Angular speed in degrees/second. The inspector clamps this between 0 and 360.

Returns

[TransformFollower](#)

This instance for fluent/chaining calls.

SetTarget(Transform)

Set the target transform to follow.

```
public TransformFollower SetTarget(Transform newTarget)
```

Parameters

newTarget Transform

The new target . Pass null to clear the target.

Returns

[TransformFollower](#)

This instance for fluent/chaining calls.

Update()

Called every frame to update the follower's transform.

```
protected virtual void Update()
```

Namespace MGeLabs.Utils.Reflection

Classes

[PredefinedAssemblyUtil](#)

Utility class for retrieving types from predefined assemblies.

Class PredefinedAssemblyUtil

Namespace: MGeLabs.Utils.[Reflection](#)

Assembly: cs.temp.dll.dll

Utility class for retrieving types from predefined assemblies.

```
public static class PredefinedAssemblyUtil
```

Inheritance

object ← PredefinedAssemblyUtil

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetTypes(Type, bool)

Gets a list of types that are subclasses of the specified generic base type.

```
public static List<Type> GetTypes(Type genericBaseType, bool includeInterfaces = false)
```

Parameters

genericBaseType Type

The generic base type to match.

includeInterfaces bool

Whether to include interfaces in the search.

Returns

List<Type>

A list of types that are subclasses of the specified generic base type.

Examples

Example usage:

```
List<Type> derivedTypes = TypeUtility.GetTypes(typeof(MyGenericBase<>));  
foreach (Type t in derivedTypes)  
{  
    Debug.Log($"Found: {t.FullName}");  
}
```

Exceptions

System.Reflection.ReflectionTypeLoadException

Thrown when there is an error loading types from an assembly.

Namespace MGeLabs.Utils.SceneManagement

Classes

[SceneLoader](#)

Manages scene loading operations in Unity.

Class SceneLoader

Namespace: MGeLabs.Utils.[SceneManagement](#)

Assembly: cs.temp.dll.dll

Manages scene loading operations in Unity.

```
public class SceneLoader : PersistentSingleton<SceneLoader>
```

Inheritance

object ← SceneLoader

Examples

```
// Example usage of SceneLoader
public class GameFlowManager : MonoBehaviour
{
    private void Start()
    {
        // Subscribe to scene loading events
        SceneLoader.Instance.OnSceneChangeStart += OnSceneStart;
        SceneLoader.Instance.OnSceneChangeProgressUpdate += OnProgressUpdate;
        SceneLoader.Instance.OnSceneChangeFinished += OnSceneFinished;

        // Start loading scene with index 2
        SceneLoader.Instance.LoadScene(2, () => Debug.Log("Scene loaded successfully!"));
    }

    private void OnSceneStart(int sceneIndex)
    {
        Debug.Log("Loading started for scene: " + sceneIndex);
    }

    private void OnProgressUpdate(float progress)
    {
        Debug.Log("Loading progress: " + (progress * 100) + "%");
    }

    private void OnSceneFinished(int sceneIndex)
    {
        Debug.Log("Loading finished for scene: " + sceneIndex);
    }
}
```



```
}  
}
```

Methods

LoadScene(int, Action)

Starts the asynchronous scene loading process.

```
public Coroutine LoadScene(int sceneIndex, Action callback = null)
```

Parameters

sceneIndex int

The index of the scene to load.

callback Action

Optional callback to invoke after the scene is loaded.

Returns

Coroutine

A Coroutine for the asynchronous operation.

Events

OnSceneChangeFinished

Event triggered when a scene change finishes.

```
public event Action<int> OnSceneChangeFinished
```

Event Type

Action<int>

OnSceneChangeProgressUpdate

Event triggered to update the progress of a scene change.

```
public event Action<float> OnSceneChangeProgressUpdate
```

Event Type

Action<float>

OnSceneChangeStart

Event triggered when a scene change starts.

```
public event Action<int> OnSceneChangeStart
```

Event Type

Action<int>